



OXTS

UCOM USER MANUAL

OXTS.COM



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Introduction

UCOM is designed to be a flexible binary data format which can be configured to output the measurements you need from your OXTS product.

In this manual you will find information on:

- + The UCOM protocol.
- + Configuring a UCOM stream.
- + The UCOM messages and signals provided by OXTS.

UCOM Overview

UCOM is broadcast over ethernet using UDP. Contained within each UCOM packet is the UCOM header, data payload, and the CRC value.

- + The header contains information about the payload as well as meta data of the UCOM stream.
- + The payload contains your configured navigation data.
- + The CRC value is a checksum to detect errors in transmission.

UCOM Header

The UCOM header contains the meta data and information about the UCOM payload and data stream. Seen in Table 1 is the UCOM header format.

Byte	Description	Information
0	Sync Characters	The sync characters are the first characters transmitted over ethernet in the UDP packet. For UCOM, they are 0x55 0x4D
1		
2	Message ID	This is a unique ID that represent messages. Messages with IDs in the range of 0-64511 are reserved for OXTS defined messages; messages with IDs in the range of 64512-65535 are reserved for user configured messages.
3		
4	Message Version	The version of a specific message
5	Trigger Type and Time Frame of the Arbitrary Time	Represented in two nibbles: + The high nibble - an enumeration of the trigger type of the packet if the packet was output on a trigger. + The low nibble - an enumeration of the time frame that the arbitrary time is represented in (bytes 6-13).
6	Arbitrary Time	Arbitrary time guaranteed to be monotonic, measured in nanoseconds, represented as a 64-bit signed integer – see UCOM Timing.
7		
8		
9		
10		
11		
12		
13		
14	Data Payload Length	Data payload length in bytes, excluding the CRC.
15		

Table 1:
The UCOM header format

UCOM Timing

UCOM provides flexible and configurable timing by using an arbitrary internal timestamp in the UCOM packet header. This timestamp is valid from start-up and is guaranteed to be monotonic. For OXTS-defined messages from an INS, the timestamp represents the strapdown navigation (SDN) time, which starts at zero and increments at each time step.

For example, on an INS configured with an IMU rate of 100 Hz, the SDN time will increase by 10 ms per step, representing the total elapsed time since start-up. The header time is stored as a 64-bit signed integer, with negative values indicating invalid time. Any packet containing an invalid timestamp should be discarded. All supported time frames are listed featured in the `oxts.dbu` file, see the Auxiliary files section.

To facilitate alignment between SDN time and other configured time datums, such as GNSS or PTP, UCOM provides a time offset message (message ID 100). The SDN time offset in the message payload represents the difference, in nanoseconds, between the SDN time in the header and a configured time datum. By reading this message, other time datums can be accurately synchronised to SDN time, ensuring that all UCOM messages remain temporally consistent.

UCOM Payload Key Components

The payload consists of a UCOM message, each message contains an array of signals. These signals represent a measurement that is able to be output. Each signal is part of a source which is a group of signals which have been naturally grouped due to the category of their output and output rates.

Messages

Messages are an array of signals which contain the configured output data. The size of a message can be any number of bytes; however, we limit the size of OXTS messages to 1452 bytes to avoid fragmentation issues caused by the default Maximum Transmission Unit (MTU) of 1500 bytes for Ethernet as it is likely to be transmitted over an IPv4 network. Limiting to 1452 bytes leaves room for the 20-byte IPv4 header, the 8-byte UDP header, the 16-byte UCOM header, and the 4-byte CRC value.

Custom Messages

Custom messages are those defined by you, the user. They can have any timing value within the header supported by the OXTS product and can have any number of signals from any source as defined by you. Currently, however, individual tinkering of the signals is not implemented. Therefore, changing the individual signal's data type, offset, scale factor, or unit will not have any effect on the final signal output.

Custom messages are also not restricted to a maximum size so you can configure your message to any size you would like. Keep in mind that the UDP protocol may split these packets up if too large as mentioned above.

Sources and Signals

A source is a natural grouping of signals, organised according to the type of measurement they represent and their inherent output rate. Sources provide a means to structure and manage signals effectively, meaning that messages can be constructed efficiently, reducing redundant data transmission and ensuring all related measurements are synchronised. A signal represents an individual measurement. By knowing which source a signal belongs to, it becomes easier to design messages that are consistent and logically grouped, ensuring that all included signals are updated simultaneously and originate from the same underlying system component.

IDs

Each signal, source, and message each contain a unique ID value. For signals and sources, these are strings that uniquely identify each entity. An example for a signal ID for the angular rate about the x-axis is, as a signal, Wx. The Source IDs contain the OXTS software responsible for this source and the abbreviated source name separated by an underscore (`\_`), e.g., BNS_SDN represents the strapdown navigator (SDN) measurements which originates from the blended navigation system (BNS). The exceptions to this rule are the TIME and META sources which do not necessarily originate from an OXTS software but are a part of the system or are there to serve basic functionality. Message IDs are integers, ranging from 0 to 64511 for OXTS defined messages, and from 64512 to 65535 for user configured messages.

CRC Value

For the cyclic redundancy check (CRC), we use a CRC-32 implementation using the polynomial 0x04C11DB7 and a 1KiB look-up table (see Appendix A); the check value is an unsigned 32-bit integer. The check value is inclusive of the header of the message too, as it contains important time information, and it is appended to the end of the UCOM packet. To calculate the check value for your packet, follow the Python code below in Figure 1 passing in the UCOM packet (without the final 4-bytes as this is the checksum),

```
def compute_crc32_value(packet):  
    """  
    packet - the raw binary data packet excluding the CRC32 value  
  
    Assuming that the look-up table is stored in a list called `crc32_lut`  
    """  
    crc = 0  
  
    for byte in packet:  
        # Bit mask to ensure valid index range in look-up table  
        index = (crc ^ byte) & 0xFF  
  
        look_up_value = crc32_lut[index]  
  
        crc = look_up_value ^ (crc >> 8)  
  
    # Ensure lower 32-bits are returned in Python so apply bit mask  
    return (~crc) & 0xFFFFFFFF
```

Figure 1:
Python function to compute the CRC-32 value which can be used to verify the integrity of a UCOM packet.

Auxiliary files

After installing the latest compatible firmware on your INS, you will be able to view two files on the INS' FTP server: oxts.dbu and oxts.dbs. These files contain read-only data in JSON format and provide you with comprehensive information available for UCOM outputs specific to your INS.

The `oxts.dbs` file contains all the possible signals in their sources along with their natural update rates, their natural types and units, as well as comprehensive names and descriptions. The `oxts.dbu` file contains all the possible OXTS defined UCOM messages possible for output, each containing the default output type and rate, in addition to the signals contained in the message and how they are encoded. The `oxts.dbu` file also contains the header timing and trigger values with their corresponding meanings in an enumeration-description object which will allow you to decode the nibbles in byte-5 of the header.

These files can be modified but are never read-in by the firmware; they are also rewritten each time the INS is power-cycled.

Example: oxts.dbu

Figure 2 shows an excerpt from the oxts.dbu file with the first message, the heartbeat, alongside key-value pairs describing the file. Each message in the file follows this format. From these messages, you can extract all relevant details, including default message information. Output frequency decimation indicates how often a message is sent relative to its natural frequency. For example, if the message has its frequency output type as IMU and has a decimation of 1, it outputs at the IMU's set rate. Higher decimation values reduce the rate; e.g., for an IMU at 100 Hz with a message having a decimation of 5 outputs at 20 Hz.

```
{
  "$schema": "DBU.schema.json",
  "DBUDescription": "The OXTS DBU with all the OXTS defined messages",
  "DBUID": "OXTS_DBU",
  "DBUName": "2026-01-12",
  "UCOMVersion": 1,
  "PossibleTriggerTypes": {
    "0": "No Trigger",
    "1": "In 1 Down",
    "2": "In 1 Up",
    "3": "Out 1",
    "4": "In 2 Down",
    "5": "In 2 Up",
    "6": "Unknown 1",
    "7": "Unknown 2",
    "8": "Out 2"
  },
  "PossibleHeaderTiming": {
    "0": "SDN",
    "1": "GNSS"
  },
  "Messages": [
    {
      "FrequencyOutputType": "IMU",
      "IntroducedInVersion": 0,
      "MessageDescription": "Heartbeat message with meta information about the
INS",
      "MessageEnabled": true,
      "MessageID": 0,
      "MessageName": "Heartbeat",
      "MessageSchemaVersion": 2,
      "MessageTiming": "SDN",
      "MessageVersion": 0,
      "OutputFrequencyDecimation": 1,
      "OutputType": "FrequencyBased",
      "SignalsInMessage": [
        {
          "SignalID": "SerialNumber",
          "SourceID": "META",
          "DataType": "U32",
          "Offset": 0.0,
          "ScaleFactor": 1.0,
          "Unit": "-"
        }
      ]
    }
  ]
}
```

Figure 2:
An excerpt of the oxts.dbu file showing the heartbeat message.

Example: oxts.dbs

Shown in Figure 3 is an excerpt of the `oxts.dbs` file. It details the signals in the time stamp source, which only contains the GNSS time stamp in this example. Here, and for all signals, you will find the native type, the native unit, and the native unit magnitude that the signals use when being encoded. All the OXTS defined messages use the native values of the signals. For any signals which represents an enumeration, the enumeration values are included within the signal object.

```
{
  "$schema": "DBS.schema.json",
  "DBSDescription": "The available signals that are available to be output",
  "DBSID": "OXTS_DBS",
  "DBSName": "OXTS DBS",
  "UCOMVersion": 1,
  "Sources": [
    {
      "SourceID": "TIME",
      "SourceName": "Time stamp source",
      "SourceDescription": "The resolved time stamps signals",
      "SignalsInSource": [
        {
          "SignalID": "Nano",
          "SignalName": "GNSS Time",
          "SignalDescription": "The time according to the GNSS sources",
          "Category": "Time",
          "NativeType": "S64",
          "NativeUnit": "s",
          "UnitMagnitude": "n",
          "IntroducedInVersion": 0
        }
      ]
    }
  ]
}
```

Figure 3:
An excerpt of the `oxts.dbs` file showing the time stamp source.

JSON schemas

Apparent in both of the previous examples is the appearance of the `$schema` keyword which references an internal schema in the firmware to verify the contents of these files and ensures that they are valid for UCOM. In this current release, these schemas are unavailable to view; however, expect to be able to view them in subsequent releases.

Backward compatibility and versioning

In the first release of UCOM (UCOM version 0), there were multiple values which determined the version: `DBUSchemaVersion`, `DBSSchemaVersion`, `DBUVersion`, and `DBSVersion`. As it was the initial release, they are all 0. Since then, they have all been encompassed into one value: the `UCOMVersion`. The following firmware are the only ones to use the old versioning system: ADK Dev ID: 250502ez, 250606ez; Device ID: 250612x3, 250502x4, 250624x1, 250642x2, 250624x3, 250624x4; NavSuite software 3.14.x.

Herein, when mentioning the UCOM version, we refer to the number that this object represents (or in the case of the initial version, version 0, the number that the four objects represent). Later in this

document, where the UCOM messages are listed, we list which version that the message was introduced in.

New firmware versions can still accept and parse the old the `mobile.dbu` configuration file, if there are any issues with that file then the validation system will output any diagnostics accordingly.

Creating a UCOM Stream

UCOM can be configured using NAVconfig which is strongly recommended. However, the individual messages can be configured by modifying the `mobile.dbu` file; we do not recommend this approach as it introduces possible errors into the JSON.

UCOM is not feature code locked, however some of the signals in the messages are feature code locked and will not be available unless you have the appropriate feature code enabled. If you have a feature code locked signal configured, you will see a quiet nan value output (conforming to the IEEE 754 standard).

Once configured, UCOM is output over Ethernet using UDP on port 50487. The sync characters which begin the UCOM header are 0x55 0x4D.

Using NAVconfig

You will find UCOM in the interfaces tab in NAVconfig. Tick the “Enable UCOM” box, then follow the configuration steps for each message you wish to enable and configure. Continue by pressing “Next”. Once the configuration has been committed to the INS, you will have enabled and configured UCOM.

Output types

UCOM messages can be output on three different conditions: output based on frequency, output based on a trigger, and output based on a changed signal.

Frequency based outputs

Frequency based output is the simplest: a message is output at a certain frequency based on the chosen, configured rate and the decimation for that message in the `mobile.dbu` file.

Trigger based outputs

When configuring a message, you can set the output to the triggers selected in NAVconfig which will output the configured UCOM messages to be output on those triggers. You can output any message on any trigger within the bounds of the INS configuration. The available trigger output types are also available in the `oxts.dbu` file.

On changed outputs

When a message is configured to be on changed, the message is only output when there is fresh data in at least one of the signals contained within the message. The only signal not to be checked is the Nano signal as it's continually incrementing.

The mobile.dbu file

UCOM uses the `mobile.dbu` file to encode the output stream. This file defines the messages and signals that are transmitted. It has the same layout as the `oxts.dbu` file and is JSON-formatted and schema-conformant. However, the `mobile.dbu` file fully defines the configured UCOM stream. As a result, it can also be used to fully decode a UCOM stream (see [Decoding a UCOM stream](#)).

Modifying the mobile.dbu file

Directly modifying the `mobile.dbu` file is not recommended, as any invalid JSON syntax will fail to configure a UCOM stream. However, the `MessageEnabled` flag within each message may be set to `true` or `false` to enable or disable message outputs as required. Changes made to OXTS defined UCOM messages, such as modifying their signals, will not affect the output of those messages. These messages are defined by OXTS and are always output in the format specified in the `oxts.dbu` file.

Dealing with invalid JSON

If it becomes necessary to edit the raw JSON within the `mobile.dbu` file, the file must remain syntactically valid. If invalid JSON is detected, no configured messages will be output. Instead, a special error DBU message (ID 64511) will be transmitted. This message contains the number of errors followed by the corresponding error messages encoded as strings. In the case of invalid JSON syntax, only a single error will be reported. The exact location of the syntax error within the file cannot currently be provided. For this reason, manual editing of the JSON is strongly discouraged, and the use of NAVconfig is highly recommended.

Further details on this error message can be found in the [OXTS Defined Messages](#) documentation later in this manual.

Diagnostics

Even if the JSON syntax is valid, issues may still occur when modifying the `mobile.dbu` file directly. While values may be syntactically correct, they may not be valid for UCOM. In addition, required fields may be missing. For example, a `DataType` must be defined for each signal in order for it to be encoded correctly. If invalid values or missing required fields are detected, the special error message (ID 64511) will again be output in place of any configured messages. Where multiple errors exist, they will be separated by commas. For less critical issues, a warning message (ID 64510) will be output. This message provides warnings about the UCOM configuration, such as the use of unknown message IDs within the OXTS message ID range.

Recommendations and limitations

While UCOM is incredibly configurable, we recommend the following when configuring UCOM:

- + The output rate for a message is set to a maximum of 100Hz
- + The total number of bytes of all UCOM messages configured should be less than 600 bytes (around 10 messages)
- + That CAN and UCOM are not enabled to be output at the same time
- + The time offset message (ID 100) cannot be configured to be output on anything other than on changed; the time offset signal it contains cannot be included in custom messages.

Note that it's possible to exceed those recommendations; however, you might notice a decrease in performance.

Decoding a UCOM stream

Each of OXTS' defined messages' formats are documented in the OXTS Defined Messages section and detail the raw binary data layout contained within a packet.

To see an example of a UCOM decoder, visit our GitHub page:

<https://github.com/OxfordTechnicalSolutions/ucom-decoder>

Data Types

Type	Data Length	Value when invalid or feature code locked
U8	Unsigned 8-bit integer	0xFF
S8	Signed 8-bit integer	0x7F
U16	Unsigned 16-bit integer	0xFFFF
S16	Signed 16-bit integer	0x7FFF
U32	Unsigned 32-bit integer	0xFFFFFFFF
S32	Signed 32-bit integer	0x7FFFFFFF
U64	Unsigned 64-bit integer	0xFFFFFFFFFFFFFFFF
S64	Signed 64-bit integer	0x7FFFFFFFFFFFFFFF
F32	32-bit Floating point number	0x7FC00000
F64	64-bit Floating point number	0x7FF8000000000000
EnS64	Combined unsigned 8-bit integer and a 64-bit signed integer; 72-bits total	0xFF7FFFFFFFFFFFFFFF
STR	Unknown until encoded; always null terminated	A null terminated string of 0s with the same length of the string encoded if not invalid

Table 2:
UCOM data types

Enumeration Plus Value Data Types

The EnS64 data type is a combination of an unsigned 8-bit integer enumeration value and a 64-bit signed integer value. The enumeration describes how the value that follows in the signal should be interpreted (see specific messages for their concrete implementation). Seen in Table 3 is an example of the SDNOFF signal in message ID 100.

Bit	Description
0-7	Unsigned 8-bit integer: Value indicating which time datum the offset is measured from relative to the SDN time datum
8-71	Signed 64-bit integer: Value of the offset

Table 3:
The bit layout for the SDNOFF signal

Compatibility

The presence of the `oxts.dbu` and `oxts.dbs` files on the INS' FTP server indicate that the INS supports UCOM. You can check the UCOM version of the files by inspecting them. The first firmware and software compatibility versions are shown in Table 4 and Table 5 respectively.

Firmware	Version
ADK Dev ID	250502ez and above
Device ID	260527x1, 260527x2, 250612x3, 250502x4, and above

Table 4:
Firmware compatibility

Software Release	Version
NavSuite	3.14.x and above

Table 5:
Software compatibility

OXTS Defined Messages and Sources

ID	Version	Name	Payload Size (bytes)
0	0	Heartbeat	4
100	0	The SDN time offsets	9
54	0	INS Status	1
1	1	OXTS navigation frame PVA	72
2	1	ISO-8855 Earth fixed frame VA	48
3	1	OXTS navigation frame PVA accuracies	80
4	1	ISO-8855 Earth fixed VA accuracies	48
5	1	GNSS Status	16
6	1	OXTS horizontal frame velocity	24
7	1	OXTS vehicle frame velocity	24
8	1	ISO-8855 intermediate frame velocity	24
9	1	ISO-8855 vehicle system velocity	24
10	1	OXTS vehicle frame linear accelerations and angular rates	48
11	1	OXTS navigation frame linear accelerations and angular rates	48
12	1	OXTS intermediate frame linear accelerations and angular rates	48
13	1	ISO-8855 Earth fixed linear accelerations and angular rates	48
14	1	ISO-8855 intermediate frame linear accelerations and angular rates	48
15	1	ISO-8855 vehicle frame linear accelerations and angular rates	48
16	1	OXTS vehicle frame filtered inertial measurements	96
17	1	OXTS navigation frame filtered inertial measurements	96

ID	Version	Name	Payload Size (bytes)
18	1	OXTS intermediate frame filtered inertial measurements	96
19	1	ISO-8855 Earth fixed filtered inertial measurements	96
20	1	ISO-8855 intermediate frame filtered inertial measurements	96
21	1	ISO-8855 vehicle frame filtered inertial measurements	96
22	1	Point of interest 1	96
23	1	Point of interest 2	96
24	1	Point of interest 3	96
25	1	Point of interest 4	96
26	1	Point of interest 5	96
27	1	Point of interest 6	96
28	1	Point of interest 7	96
29	1	Point of interest 8	96
30	1	Distance and speed	48
31	1	Track, slip, curvature, angle gradient	32
32	1	Heave	8
33	1	Primary GNSS card measurements	48
34	1	Secondary GNSS card measurements	24
35	1	Local coordinate PVA	48
36	0	GAD decoder information	8
37	0	GAD statuses	4
38	0	Position aiding innovations	28
39	0	Velocity aiding innovations	28
40	0	Attitude aiding innovations	20
41	0	GAD packet statistics	14
42	0	Position aiding in the INS frame	28
43	0	Velocity aiding of the INS in navigation frame	29
44	0	Orientation aiding of the INS in dual-antenna frame	20

ID	Version	Name	Payload Size (bytes)
45	0	Covariances of the latest position aiding update in navigation frame	52
46	0	Covariances of the latest velocity aiding update in navigation frame	53
47	0	Raw IMU measurements	48
48	0	OXTS vehicle frame linear jerks and angular accelerations	48
49	0	OXTS navigation frame linear jerks and angular accelerations	48
50	0	OXTS intermediate frame linear jerks and angular accelerations	48
51	0	ISO-8855 Earth fixed linear jerks and angular accelerations	48
52	0	ISO-8855 intermediate frame linear jerks and angular accelerations	48
53	0	ISO-8855 vehicle frame linear jerks and angular accelerations	48
64510	0	Warnings	Unknown
64511	0	Invalid mobile DBU	Unknown

Table 6:
Messages

ID	Name	Number Of Signals
TIME	Time stamp source	4
META	Meta source	11
BNS_STS	BNS status source	5
BNS_SDN	Strapdown navigator source	54
BNS_FLT	Filtered blended navigation source	32
BNS_DER	Derived signals source	110
BNS_ISO	ISO-8855 source	48
BNS_FLTISO	ISO-8855 filtered source	36
BNS_ACC	Blended navigation system accuracies source	10
BNS_ACCISO	Blended navigation system ISO-8855 frame accuracies source	6
BNS_LOC	Local values source	12
GNSS_STS	GNSS status source	76
GNSS_PGC	Primary GNSS card source	24
GNSS_SGC	Secondary GNSS card source	18
AID_INNOV	Innovations	39
INS_CFG	INS Configuration source	70
BNS_CFG	BNS Configuration source	51
GNSS_CFG	GNSS Configuration source	4
BNS_OPT	Blended navigation system optimised parameters source	7
BNS_ACCOPT	Blended navigation system optimised parameter accuracies source	7
BNS_TRG	Blended navigation trigger source	10
BNS_STATS	Blended navigation statistics source	24
BNS_ACCDER	Blended navigation system accuracies for derived signals source	4
BNS_BIAS	The biases for different sensors in the blended navigation system source	12
BNS_ACCBIAS	The accuracies of biases of sensors in the blended navigation system	12

ID	Name	Number Of Signals
GNSS_STATS	Statistics for the GNSS engine source	6
BASE_STS	Base station status source	2
BNS_GAD	Generic aiding	11
BNS_WSP	Wheel speed source	17
BNS_CAM	Camera source	2
GNSS_ACC	GNSS engine accuracies	3
GNSS_EGC	External GNSS card source	16

Table 7:
Sources

OXTS Defined Messages

ID: 0		Heartbeat		
Version		0		
Description		Heartbeat message with meta information about the INS		
Header Timing		SDN		
Payload Size		4 Bytes		
Introduced In		UCOM Version 0		
Offset	Type	Signal ID	Name	Unit
0	U32	SerialNumber	Serial number	-

Table 8:
Heartbeat Message

ID: 100		The SDN time offsets		
Version		0		
Description		The time offsets from the SDN time in the header to the time in the message payload described by the enum value: byte 1 of the payload is the enum value and bytes 2-9 are the offset value		
Header Timing		SDN		
Payload Size		9 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	EnS6 4	SDNOFF	SDN Time Offset	ns

Table 9:
The SDN time offsets Message

ID: 54		INS Status		
Version		0		
Description		The current INS navigation status from the blended navigation system		
Header Timing		SDN		
Payload Size		1 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	U8	InsNavMode	Navigation Status	-

Table 10:
INS Status Message

ID: 1		OXTS navigation frame PVA		
Version		1		
Description		Position, velocity, and attitude in OXTS navigation frame (NED) in the configured output datum		
Header Timing		SDN		
Payload Size		72 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	Lat	Latitude	deg
8	F64	Lon	Longitude	deg
16	F64	Alt	Altitude	m
24	F64	Vn	Velocity North	m/s
32	F64	Ve	Velocity East	m/s
40	F64	Vd	Velocity down	m/s
48	F64	Heading	Heading	deg
56	F64	Pitch	Pitch	deg
64	F64	Roll	Roll	deg

Table 11:
OXTS navigation frame PVA Message

ID: 2		ISO-8855 Earth fixed frame VA		
Version		1		
Description		Velocity and attitude in the ISO 8855 Earth fixed frame (ENU)		
Header Timing		SDN		
Payload Size		48 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	IsoVnX	ISO e.f.s. east velocity	m/s
8	F64	IsoVnY	ISO e.f.s. north velocity	m/s
16	F64	IsoVnZ	ISO e.f.s. vertical velocity	m/s
24	F64	IsoYaw	ISO yaw angle	deg
32	F64	IsoPitch	ISO pitch angle	deg
40	F64	IsoRoll	ISO roll angle	deg

Table 12:
ISO-8855 Earth fixed frame VA Message

ID: 3		OXTS navigation frame PVA accuracies		
Version		1		
Description		OXTS navigation frame (NED) position, velocity, and attitude accuracies - these accuracies are taken from the Kalman filter and do not account for output smoothing		
Header Timing		SDN		
Payload Size		80 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	NorthAcc	North accuracy	m
8	F64	EastAcc	East accuracy	m
16	F64	AltAcc	Position accuracy down	m
24	F64	VnAcc	Velocity North accuracy	m/s
32	F64	VeAcc	Velocity East accuracy	m/s
40	F64	VdAcc	Velocity down accuracy	m/s
48	F64	VuAcc	Velocity up accuracy	m/s
56	F64	HeadingAcc	Heading accuracy	deg
64	F64	PitchAcc	Pitch accuracy	deg
72	F64	RollAcc	Roll accuracy	deg

Table 13:
OXTS navigation frame PVA accuracies Message

ID: 4		ISO-8855 Earth fixed VA accuracies		
Version		1		
Description		ISO-8855 Earth fixed (ENU) velocity and attitude accuracies - these accuracies are taken from the Kalman filter and do not account for output smoothing		
Header Timing		SDN		
Payload Size		48 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	IsoVnXAcc	ENU East velocity accuracy	m/s
8	F64	IsoVnYAcc	ENU North velocity accuracy	m/s
16	F64	IsoVnZAcc	ENU vertical velocity accuracy	m/s
24	F64	IsoYawAcc	Yaw accuracy	deg
32	F64	IsoPitchAcc	Pitch accuracy	deg
40	F64	IsoRollAcc	Roll accuracy	deg

Table 14:
ISO-8855 Earth fixed VA accuracies Message

ID: 5		GNSS Status		
Version		1		
Description		GNSS Status		
Header Timing		SDN		
Payload Size		16 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	U8	GnssPosMode	GNSS position mode	-
1	U8	GnssVelMode	GNSS velocity mode	-
2	U8	GnssAttMode	GNSS dual antenna attitude mode	-
3	F64	GnssDiffAge	Differential GNSS age	s
11	U8	PriSatTrac	Primary satellites tracked	-
12	U8	SecSatTrac	Secondary satellites tracked	-
13	U8	GnssPosNumSats	Number of GNSS satellites used	-
14	U8	HeaSatTrac	Satellites tracked for attitude	-
15	U8	GnssAttNumSats	GnssAttNumSats	-

Table 15:
GNSS Status Message

ID: 6		OXTS horizontal frame velocity		
Version		1		
Description		OXTS horizontal frame (level frame; FLD) liner velocities		
Header Timing		SDN		
Payload Size		24 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	Vf	Velocity forward	m/s
8	F64	VI	Velocity lateral	m/s
16	F64	Vd	Velocity down	m/s

Table 16:
OXTS horizontal frame velocity Message

ID: 7		OXTS vehicle frame velocity		
Version		1		
Description		OXTS vehicle system frame (XYZ) liner velocities		
Header Timing		SDN		
Payload Size		24 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	Vx	X Velocity	m/s
8	F64	Vy	Y Velocity	m/s
16	F64	Vz	Z Velocity	m/s

Table 17:
OXTS vehicle frame velocity Message

ID: 8		ISO-8855 intermediate frame velocity		
Version		1		
Description		ISO-8855 intermediate frame (FLU) liner velocities		
Header Timing		SDN		
Payload Size		24 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	IsoVhX	ISO i.s. longitudinal velocity	m/s
8	F64	IsoVhY	ISO i.s. lateral velocity	m/s
16	F64	IsoVhZ	ISO i.s. vertical velocity	m/s

Table 18:
ISO-8855 intermediate frame velocity Message

ID: 9		ISO-8855 vehicle system velocity		
Version		1		
Description		ISO 8855 vehicle system frame (XYZ) liner velocities		
Header Timing		SDN		
Payload Size		24 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	IsoVoX	ISO v.s. longitudinal velocity	m/s
8	F64	IsoVoY	ISO v.s. lateral velocity	m/s
16	F64	IsoVoZ	ISO v.s. vertical velocity	m/s

Table 19:
ISO-8855 vehicle system velocity Message

ID: 10		OXTS vehicle frame linear accelerations and angular rates		
Version		1		
Description		Linear accelerations and angular rates in OXTS vehicle frame (XYZ) - the Kalman filter model has been applied to these signals		
Header Timing		SDN		
Payload Size		48 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	Ax	X Acceleration	m/s ²
8	F64	Ay	Y Acceleration	m/s ²
16	F64	Az	Z Acceleration	m/s ²
24	F64	Wx	X Angular rate	deg/s
32	F64	Wy	Y Angular rate	deg/s
40	F64	Wz	Z Angular rate	deg/s

Table 20:
OXTS vehicle frame linear accelerations and angular rates Message

ID: 11		OXTS navigation frame linear accelerations and angular rates		
Version		1		
Description		Linear accelerations and angular rates in OXTS navigation frame (NED) - the Kalman filter model has been applied to these signals		
Header Timing		SDN		
Payload Size		48 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	An	Acceleration North	m/s ²
8	F64	Ae	Acceleration East	m/s ²
16	F64	Ad	Acceleration down	m/s ²
24	F64	Wn	Angular rate North	deg/s
32	F64	We	Angular rate East	deg/s
40	F64	Wd	Angular rate down	deg/s

Table 21:
OXTS navigation frame linear accelerations and angular rates Message

ID: 12		OXTS intermediate frame linear accelerations and angular rates		
Version		1		
Description		Linear accelerations and angular rates in OXTS intermediate frame (FLD) - the Kalman filter model has been applied to these signals		
Header Timing		SDN		
Payload Size		48 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	Af	Acceleration forward	m/s^2
8	F64	Al	Acceleration lateral	m/s^2
16	F64	Ad	Acceleration down	m/s^2
24	F64	Wf	Angular rate forward	deg/s
32	F64	Wl	Angular rate lateral	deg/s
40	F64	Wd	Angular rate down	deg/s

Table 22:
OXTS intermediate frame linear accelerations and angular rates Message

ID: 13		ISO-8855 Earth fixed linear accelerations and angular rates		
Version		1		
Description		ISO-8855 Earth Fixed (ENU) inertial measurement - the Kalman filter model has been applied to these signals		
Header Timing		SDN		
Payload Size		48 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	IsoAnX	ISO e.f.s. east acceleration	m/s ²
8	F64	IsoAnY	ISO e.f.s. north acceleration	m/s ²
16	F64	IsoAnZ	ISO e.f.s. vertical acceleration	m/s ²
24	F64	IsoWnX	ISO e.f.s. roll velocity	deg/s
32	F64	IsoWnY	ISO e.f.s. pitch velocity	deg/s
40	F64	IsoWnZ	ISO e.f.s. yaw velocity	deg/s

Table 23:
ISO-8855 Earth fixed linear accelerations and angular rates Message

ID: 14		ISO-8855 intermediate frame linear accelerations and angular rates		
Version		1		
Description		ISO 8855 intermediate frame (FLD) linear accelerations and angular rates - the Kalman filter model has been applied to these signals		
Header Timing		SDN		
Payload Size		48 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	IsoAhX	ISO i.s. longitudinal acceleration	m/s^2
8	F64	IsoAhY	ISO i.s. lateral acceleration	m/s^2
16	F64	IsoAhZ	ISO i.s. vertical acceleration	m/s^2
24	F64	IsoWhX	ISO i.s. roll velocity	deg/s
32	F64	IsoWhY	ISO i.s. pitch velocity	deg/s
40	F64	IsoWhZ	ISO i.s. yaw velocity	deg/s

Table 24:
ISO-8855 intermediate frame linear accelerations and angular rates Message

ID: 15		ISO-8855 vehicle frame linear accelerations and angular rates		
Version		1		
Description		ISO 8855 vehicle frame (XYZ) linear accelerations and angular rates - the Kalman filter model has been applied to these signals		
Header Timing		SDN		
Payload Size		48 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	IsoAoX	ISO v.s. longitudinal acceleration	m/s^2
8	F64	IsoAoY	ISO v.s. lateral acceleration	m/s^2
16	F64	IsoAoZ	ISO v.s. vertical acceleration	m/s^2
24	F64	IsoWoX	ISO v.s. roll velocity	deg/s
32	F64	IsoWoY	ISO v.s. pitch velocity	deg/s
40	F64	IsoWoZ	ISO v.s. yaw velocity	deg/s

Table 25:
ISO-8855 vehicle frame linear accelerations and angular rates Message

ID: 16		OXTS vehicle frame filtered inertial measurements		
Version		1		
Description		Filtered inertial measurements in OXTS vehicle frame (XYZ) - the Kalman filter model has been applied to these signals		
Header Timing		SDN		
Payload Size		96 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	FiltAx	Filtered X Acceleration	m/s ²
8	F64	FiltAy	Filtered Y Acceleration	m/s ²
16	F64	FiltAz	Filtered Z Acceleration	m/s ²
24	F64	FiltJx	Filtered X Jerk	m/s ³
32	F64	FiltJy	Filtered Y Jerk	m/s ³
40	F64	FiltJz	Filtered Z Jerk	m/s ³
48	F64	FiltWx	Filtered X Angular rate	deg/s
56	F64	FiltWy	Filtered Y Angular rate	deg/s
64	F64	FiltWz	Filtered Z Angular rate	deg/s
72	F64	FiltYx	Filtered X Angular Acceleration	deg/s ²
80	F64	FiltYy	Filtered Y Angular Acceleration	deg/s ²
88	F64	FiltYz	Filtered Z Angular Acceleration	deg/s ²

Table 26:
OXTS vehicle frame filtered inertial measurements Message

ID: 17		OXTS navigation frame filtered inertial measurements		
Version		1		
Description		Filtered inertial measurements in OXTS navigation frame (NED) - the Kalman filter model has been applied to these signals		
Header Timing		SDN		
Payload Size		96 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	FiltAn	Filtered Acceleration North	m/s ²
8	F64	FiltAe	Filtered Acceleration East	m/s ²
16	F64	FiltAd	Filtered Acceleration down	m/s ²
24	F64	FiltJn	Filtered Jerk North	m/s ³
32	F64	FiltJe	Filtered Jerk East	m/s ³
40	F64	FiltJd	Filtered Jerk down	m/s ³
48	F64	FiltWn	Filtered Angular rate North	deg/s
56	F64	FiltWe	Filtered Angular rate East	deg/s
64	F64	FiltWd	Filtered Angular rate down	deg/s
72	F64	FiltYn	Filtered Angular acceleration North	deg/s ²
80	F64	FiltYe	Filtered Angular acceleration East	deg/s ²
88	F64	FiltYd	Filtered Angular acceleration down	deg/s ²

Table 27:
OXTS navigation frame filtered inertial measurements Message

ID: 18		OXTS intermediate frame filtered inertial measurements		
Version		1		
Description		Filtered inertial measurements in OXTS intermediate frame (FLD) - the Kalman filter model has been applied to these signals		
Header Timing		SDN		
Payload Size		96 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	FiltAf	Acceleration filtered forward	m/s ²
8	F64	FiltAl	Acceleration filtered lateral	m/s ²
16	F64	FiltAd	Filtered Acceleration down	m/s ²
24	F64	FiltJf	Jerk filtered forward	m/s ³
32	F64	FiltJl	Jerk filtered lateral	m/s ³
40	F64	FiltJd	Filtered Jerk down	m/s ³
48	F64	FiltWf	Filtered Intermediate frame angular rate forward	deg/s
56	F64	FiltWl	Filtered Intermediate frame lateral angular rate	deg/s
64	F64	FiltWd	Filtered Angular rate down	deg/s
72	F64	FiltYf	Angular acceleration filtered forward	deg/s ²
80	F64	FiltYl	Angular acceleration filtered lateral	deg/s ²
88	F64	FiltYd	Filtered Angular acceleration down	deg/s ²

Table 28:
OXTS intermediate frame filtered inertial measurements Message

ID: 19		ISO-8855 Earth fixed filtered inertial measurements		
Version		1		
Description		ISO-8855 Earth Fixed (ENU) filtered inertial measurements - the Kalman filter model has been applied to these signals		
Header Timing		SDN		
Payload Size		96 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	FiltIsoAnX	ISO e.f.s. east filtered acceleration	m/s ²
8	F64	FiltIsoAnY	ISO e.f.s. north filtered acceleration	m/s ²
16	F64	FiltIsoAnZ	ISO e.f.s. vertical filtered acceleration	m/s ²
24	F64	FiltIsoJnX	ISO e.f.s. east filtered jerk	m/s ³
32	F64	FiltIsoJnY	ISO e.f.s. north filtered jerk	m/s ³
40	F64	FiltIsoJnZ	ISO e.f.s. vertical filtered jerk	m/s ³
48	F64	FiltWnX	Filtered ENU Roll Velocity	deg/s
56	F64	FiltWnY	Filtered ENU Pitch Velocity	deg/s
64	F64	FiltWnZ	Filtered ENU Yaw Velocity	deg/s
72	F64	FiltIsoYnX	ISO e.f.s. roll filtered acceleration	deg/s ²
80	F64	FiltIsoYnY	ISO e.f.s. pitch filtered acceleration	deg/s ²
88	F64	FiltIsoYnZ	ISO e.f.s. yaw filtered acceleration	deg/s ²

Table 29:
ISO-8855 Earth fixed filtered inertial measurements Message

ID: 20		ISO-8855 intermediate frame filtered inertial measurements		
Version		1		
Description		ISO 8855 intermediate frame (FLD) filtered inertial measurements - the Kalman filter model has been applied to these signals		
Header Timing		SDN		
Payload Size		96 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	FiltIsoAhX	ISO i.s. longitudinal filtered acceleration	m/s ²
8	F64	FiltIsoAhY	ISO i.s. lateral filtered acceleration	m/s ²
16	F64	FiltIsoAhZ	ISO i.s. vertical filtered acceleration	m/s ²
24	F64	FiltIsoJhX	ISO i.s. longitudinal filtered jerk	m/s ³
32	F64	FiltIsoJhY	ISO i.s. lateral filtered jerk	m/s ³
40	F64	FiltIsoJhZ	ISO i.s. vertical filtered jerk	m/s ³
48	F64	FiltWhX	Filtered Intermediate Roll Velocity	deg/s
56	F64	FiltWhY	Filtered Intermediate Pitch Velocity	deg/s
64	F64	FiltWhZ	Filtered Intermediate Yaw Velocity	deg/s
72	F64	FiltIsoYhX	ISO i.s. roll filtered acceleration	deg/s ²
80	F64	FiltIsoYhY	ISO i.s. pitch filtered acceleration	deg/s ²
88	F64	FiltIsoYhZ	ISO i.s. yaw filtered acceleration	deg/s ²

Table 30:
ISO-8855 intermediate frame filtered inertial measurements Message

ID: 21		ISO-8855 vehicle frame filtered inertial measurements		
Version		1		
Description		ISO 8855 vehicle frame (XYZ) filtered inertial measurements - the Kalman filter model has been applied to these signals		
Header Timing		SDN		
Payload Size		96 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	FiltIsoAoX	ISO v.s. longitudinal filtered acceleration	m/s ²
8	F64	FiltIsoAoY	ISO v.s. lateral filtered acceleration	m/s ²
16	F64	FiltIsoAoZ	ISO v.s. vertical filtered acceleration	m/s ²
24	F64	FiltIsoJoX	ISO v.s. longitudinal filtered jerk	m/s ³
32	F64	FiltIsoJoY	ISO v.s. lateral filtered jerk	m/s ³
40	F64	FiltIsoJoZ	ISO v.s. vertical filtered jerk	m/s ³
48	F64	FiltWoX	Filtered Vehicle System Roll Velocity	deg/s
56	F64	FiltWoY	Filtered Vehicle System Pitch Velocity	deg/s
64	F64	FiltWoZ	Filtered Vehicle System Yaw Velocity	deg/s
72	F64	FiltIsoYoX	ISO v.s. roll filtered acceleration	deg/s ²
80	F64	FiltIsoYoY	ISO v.s. pitch filtered acceleration	deg/s ²
88	F64	FiltIsoYoZ	ISO v.s. yaw filtered acceleration	deg/s ²

Table 31:
ISO-8855 vehicle frame filtered inertial measurements Message

ID: 22		Point of interest 1		
Version		1		
Description		Point of interest 1 (Linear velocity (FLD), 2D speed, accelerations (FLD and NED), track, slip, curvature)		
Header Timing		SDN		
Payload Size		96 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	MeasPt1_Vf	Intermediate frame velocity forward point 1	m/s
8	F64	MeasPt1_Vl	Intermediate frame velocity lateral point 1	m/s
16	F64	MeasPt1_Vd	Intermediate frame velocity down point 1	m/s
24	F64	MeasPt1_Speed2d	Speed 2D measurement point 1	m/s
32	F64	MeasPt1_Af	Intermediate frame acceleration forward point 1	m/s ²
40	F64	MeasPt1_Al	Intermediate frame acceleration lateral point 1	m/s ²
48	F64	MeasPt1_Ad	NED frame acceleration down point 1	m/s ²
56	F64	MeasPt1_An	NED frame acceleration north point 1	m/s ²
64	F64	MeasPt1_Ae	NED frame acceleration east point 1	m/s ²
72	F64	MeasPt1_Track	Track angle (slip point 1)	deg
80	F64	MeasPt1_Slip	Slip angle (slip point 1)	deg
88	F64	MeasPt1_Curvature	Curvature (slip point 1)	1/m

Table 32:
Point of interest 1 Message

ID: 23		Point of interest 2		
Version		1		
Description		Point of interest 2 (Linear velocity (FLD), 2D speed, accelerations (FLD and NED), track, slip, curvature)		
Header Timing		SDN		
Payload Size		96 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	MeasPt2_Vf	Intermediate frame velocity forward point 2	m/s
8	F64	MeasPt2_Vl	Intermediate frame velocity lateral point 2	m/s
16	F64	MeasPt2_Vd	Intermediate frame velocity down point 2	m/s
24	F64	MeasPt2_Speed2d	Speed 2D measurement point 2	m/s
32	F64	MeasPt2_Af	Intermediate frame acceleration forward point 2	m/s ²
40	F64	MeasPt2_Al	Intermediate frame acceleration lateral point 2	m/s ²
48	F64	MeasPt2_Ad	NED frame acceleration down point 2	m/s ²
56	F64	MeasPt2_An	NED frame acceleration north point 2	m/s ²
64	F64	MeasPt2_Ae	NED frame acceleration east point 2	m/s ²
72	F64	MeasPt2_Track	Track angle (slip point 2)	deg
80	F64	MeasPt2_Slip	Slip angle (slip point 2)	deg
88	F64	MeasPt2_Curvature	Curvature (slip point 2)	1/m

Table 33:
Point of interest 2 Message

ID: 24		Point of interest 3		
Version		1		
Description		Point of interest 3 (Linear velocity (FLD), 2D speed, accelerations (FLD and NED), track, slip, curvature)		
Header Timing		SDN		
Payload Size		96 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	MeasPt3_Vf	Intermediate frame velocity forward point 3	m/s
8	F64	MeasPt3_Vl	Intermediate frame velocity lateral point 3	m/s
16	F64	MeasPt3_Vd	Intermediate frame velocity down point 3	m/s
24	F64	MeasPt3_Speed2d	Speed 2D measurement point 3	m/s
32	F64	MeasPt3_Af	Intermediate frame acceleration forward point 3	m/s ²
40	F64	MeasPt3_Al	Intermediate frame acceleration lateral point 3	m/s ²
48	F64	MeasPt3_Ad	NED frame acceleration down point 3	m/s ²
56	F64	MeasPt3_An	NED frame acceleration north point 3	m/s ²
64	F64	MeasPt3_Ae	NED frame acceleration east point 3	m/s ²
72	F64	MeasPt3_Track	Track angle (slip point 3)	deg
80	F64	MeasPt3_Slip	Slip angle (slip point 3)	deg
88	F64	MeasPt3_Curvature	Curvature (slip point 3)	1/m

Table 34:
Point of interest 3 Message

ID: 25		Point of interest 4		
Version		1		
Description		Point of interest 4 (Linear velocity (FLD), 2D speed, accelerations (FLD and NED), track, slip, curvature)		
Header Timing		SDN		
Payload Size		96 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	MeasPt4_Vf	Intermediate frame velocity forward point 4	m/s
8	F64	MeasPt4_Vl	Intermediate frame velocity lateral point 4	m/s
16	F64	MeasPt4_Vd	Intermediate frame velocity down point 4	m/s
24	F64	MeasPt4_Speed2d	Speed 2D measurement point 4	m/s
32	F64	MeasPt4_Af	Intermediate frame acceleration forward point 4	m/s ²
40	F64	MeasPt4_Al	Intermediate frame acceleration lateral point 4	m/s ²
48	F64	MeasPt4_Ad	NED frame acceleration down point 4	m/s ²
56	F64	MeasPt4_An	NED frame acceleration north point 4	m/s ²
64	F64	MeasPt4_Ae	NED frame acceleration east point 4	m/s ²
72	F64	MeasPt4_Track	Track angle (slip point 4)	deg
80	F64	MeasPt4_Slip	Slip angle (slip point 4)	deg
88	F64	MeasPt4_Curvature	Curvature (slip point 4)	1/m

Table 35:
Point of interest 4 Message

ID: 26		Point of interest 5		
Version		1		
Description		Point of interest 5 (Linear velocity (FLD), 2D speed, accelerations (FLD and NED), track, slip, curvature)		
Header Timing		SDN		
Payload Size		96 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	MeasPt5_Vf	Intermediate frame velocity forward point 5	m/s
8	F64	MeasPt5_Vl	Intermediate frame velocity lateral point 5	m/s
16	F64	MeasPt5_Vd	Intermediate frame velocity down point 5	m/s
24	F64	MeasPt5_Speed2d	Speed 2D measurement point 5	m/s
32	F64	MeasPt5_Af	Intermediate frame acceleration forward point 5	m/s ²
40	F64	MeasPt5_Al	Intermediate frame acceleration lateral point 5	m/s ²
48	F64	MeasPt5_Ad	NED frame acceleration down point 5	m/s ²
56	F64	MeasPt5_An	NED frame acceleration north point 5	m/s ²
64	F64	MeasPt5_Ae	NED frame acceleration east point 5	m/s ²
72	F64	MeasPt5_Track	Track angle (slip point 5)	deg
80	F64	MeasPt5_Slip	Slip angle (slip point 5)	deg
88	F64	MeasPt5_Curvature	Curvature (slip point 5)	1/m

Table 36:
Point of interest 5 Message

ID: 27		Point of interest 6		
Version		1		
Description		Point of interest 6 (Linear velocity (FLD), 2D speed, accelerations (FLD and NED), track, slip, curvature)		
Header Timing		SDN		
Payload Size		96 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	MeasPt6_Vf	Intermediate frame velocity forward point 6	m/s
8	F64	MeasPt6_Vl	Intermediate frame velocity lateral point 6	m/s
16	F64	MeasPt6_Vd	Intermediate frame velocity down point 6	m/s
24	F64	MeasPt6_Speed2d	Speed 2D measurement point 6	m/s
32	F64	MeasPt6_Af	Intermediate frame acceleration forward point 6	m/s ²
40	F64	MeasPt6_Al	Intermediate frame acceleration lateral point 6	m/s ²
48	F64	MeasPt6_Ad	NED frame acceleration down point 6	m/s ²
56	F64	MeasPt6_An	NED frame acceleration north point 6	m/s ²
64	F64	MeasPt6_Ae	NED frame acceleration east point 6	m/s ²
72	F64	MeasPt6_Track	Track angle (slip point 6)	deg
80	F64	MeasPt6_Slip	Slip angle (slip point 6)	deg
88	F64	MeasPt6_Curvature	Curvature (slip point 6)	1/m

Table 37:
Point of interest 6 Message

ID: 28		Point of interest 7		
Version		1		
Description		Point of interest 7 (Linear velocity (FLD), 2D speed, accelerations (FLD and NED), track, slip, curvature)		
Header Timing		SDN		
Payload Size		96 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	MeasPt7_Vf	Intermediate frame velocity forward point 7	m/s
8	F64	MeasPt7_Vl	Intermediate frame velocity lateral point 7	m/s
16	F64	MeasPt7_Vd	Intermediate frame velocity down point 7	m/s
24	F64	MeasPt7_Speed2d	Speed 2D measurement point 7	m/s
32	F64	MeasPt7_Af	Intermediate frame acceleration forward point 7	m/s ²
40	F64	MeasPt7_Al	Intermediate frame acceleration lateral point 7	m/s ²
48	F64	MeasPt7_Ad	NED frame acceleration down point 7	m/s ²
56	F64	MeasPt7_An	NED frame acceleration north point 7	m/s ²
64	F64	MeasPt7_Ae	NED frame acceleration east point 7	m/s ²
72	F64	MeasPt7_Track	Track angle (slip point 7)	deg
80	F64	MeasPt7_Slip	Slip angle (slip point 7)	deg
88	F64	MeasPt7_Curvature	Curvature (slip point 7)	1/m

Table 38:
Point of interest 7 Message

ID: 29		Point of interest 8		
Version		1		
Description		Point of interest 8 (Linear velocity (FLD), 2D speed, accelerations (FLD and NED), track, slip, curvature)		
Header Timing		SDN		
Payload Size		96 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	MeasPt8_Vf	Intermediate frame velocity forward point 8	m/s
8	F64	MeasPt8_Vl	Intermediate frame velocity lateral point 8	m/s
16	F64	MeasPt8_Vd	Intermediate frame velocity down point 8	m/s
24	F64	MeasPt8_Speed2d	Speed 2D measurement point 8	m/s
32	F64	MeasPt8_Af	Intermediate frame acceleration forward point 8	m/s ²
40	F64	MeasPt8_Al	Intermediate frame acceleration lateral point 8	m/s ²
48	F64	MeasPt8_Ad	NED frame acceleration down point 8	m/s ²
56	F64	MeasPt8_An	NED frame acceleration north point 8	m/s ²
64	F64	MeasPt8_Ae	NED frame acceleration east point 8	m/s ²
72	F64	MeasPt8_Track	Track angle (slip point 8)	deg
80	F64	MeasPt8_Slip	Slip angle (slip point 8)	deg
88	F64	MeasPt8_Curvature	Curvature (slip point 8)	1/m

Table 39:
Point of interest 8 Message

ID: 30		Distance and speed		
Version		1		
Description		Distance travelled and speed (2D and 3D)		
Header Timing		SDN		
Payload Size		48 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	Dist2d	Distance horizontal	m
8	F64	Dist2dHold	Distance horizontal with hold	m
16	F64	Dist3d	Distance 3D	m
24	F64	Dist3dHold	Distance 3D with hold	m
32	F64	Speed2d	Speed horizontal	m/s
40	F64	Speed3d	Speed 3D	m/s

Table 40:
Distance and speed Message

ID: 31		Track, slip, curvature, angle gradient		
Version		1		
Description		Track, slip, curvature, and angle gradient		
Header Timing		SDN		
Payload Size		32 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	Curvature	Curvature	1/m
8	F64	Slip	Slip angle	deg
16	F64	Track	Track angle	deg
24	F64	AngleGradient	Angle Gradient	deg

Table 41:
Track, slip, curvature, angle gradient Message

ID: 32		Heave		
Version		1		
Description		Heave		
Header Timing		SDN		
Payload Size		8 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	Heave	Heave	m

Table 42:
Heave Message

ID: 33		Primary GNSS card measurements		
Version		1		
Description		Primary GNSS card native measurements		
Header Timing		SDN		
Payload Size		48 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	PGCApproxLat	Primary GNSS card latitude	-
8	F64	PGCApproxLon	Primary GNSS card longitude	deg
16	F64	PGCApproxAlt	Primary GNSS card altitude	m
24	F64	PGCVn	Primary GNSS card velocity North	m/s
32	F64	PGCVe	Primary GNSS card velocity East	m/s
40	F64	PGCVd	Primary GNSS card velocity down	m/s

Table 43:
Primary GNSS card measurements Message

ID: 34		Secondary GNSS card measurements		
Version		1		
Description		Secondary GNSS card native measurements		
Header Timing		SDN		
Payload Size		24 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	SGCAproxLat	Secondary GNSS card latitude	deg
8	F64	SGCAproxLon	Secondary GNSS card longitude	deg
16	F64	SGCAproxAlt	Secondary GNSS card altitude	m

Table 44:
Secondary GNSS card measurements Message

ID: 35		Local coordinate PVA		
Version		1		
Description		Local coordinate position, velocity, and attitude		
Header Timing		SDN		
Payload Size		48 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	RefFrameX	Local co-ordinates position Xr	m
8	F64	RefFrameY	Local co-ordinates position Yr	m
16	F64	RefFrameVelX	Local co-ordinates velocity Xr	m/s
24	F64	RefFrameVelY	Local co-ordinates velocity Yr	m/s
32	F64	RefFrameYaw	Local co-ordinates yaw angle	deg
40	F64	RefFrameTrack	Local co-ordinates track angle	deg

Table 45:
Local coordinate PVA Message

ID: 36		GAD decoder information		
Version		0		
Description		GAD decoder information		
Header Timing		SDN		
Payload Size		8 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	U32	GADChars	GAD characters received	-
4	U32	GADPkts	GAD packets decoded	-

Table 46:
GAD decoder information Message

ID: 37		GAD statuses		
Version		0		
Description		GAD aiding statuses		
Header Timing		SDN		
Payload Size		4 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	U8	GADPosStatus	GAD position status	-
1	U8	GADVelStatus	GAD velocity status	-
2	U8	GADAttStatus	GAD attitude status	-
3	U8	GADAngRateS tatus	GAD angular rate status	-

Table 47:
GAD statuses Message

ID: 38		Position aiding innovations		
Version		0		
Description		Position aiding innovations		
Header Timing		SDN		
Payload Size		28 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	S32	PosAidingStreamID	Position Aiding stream ID	-
4	F64	InnPosX	Innovation position north	-
12	F64	InnPosY	Innovation position east	-
20	F64	InnPosZ	Innovation position down	-

Table 48:
Position aiding innovations Message

ID: 39		Velocity aiding innovations		
Version		0		
Description		Velocity aiding innovations		
Header Timing		SDN		
Payload Size		28 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	S32	VelAidingStreamID	Velocity Aiding stream ID	-
4	F64	InnVelX	Innovation velocity north	-
12	F64	InnVelY	Innovation velocity east	-
20	F64	InnVelZ	Innovation velocity down	-

Table 49:
Velocity aiding innovations Message

ID: 40		Attitude aiding innovations		
Version		0		
Description		Attitude aiding innovations		
Header Timing		SDN		
Payload Size		20 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	S32	AttAidingStreamID	Attitude Aiding stream ID	-
4	F64	InnPitch	Innovation pitch	-
12	F64	InnHeading	Innovation heading	-

Table 50:
Attitude aiding innovations Message

ID: 41		GAD packet statistics		
Version		0		
Description		GAD packet statistics		
Header Timing		SDN		
Payload Size		14 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	U8	GADLatestStreamID	GAD latest stream ID	-
1	U32	GADNumEarly	GAD Early Packets	-
5	U32	GADNumLate	GAD Late Packets	-
9	U32	GADNumScheduled	GAD Scheduled Packets	-
13	U8	GADLatestStatus	GAD Latest Packet Status	-

Table 51:
GAD packet statistics Message

ID: 42		Position aiding in the INS frame		
Version		0		
Description		Position aiding update of INS as WGS84		
Header Timing		SDN		
Payload Size		28 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	S32	PosAidingStreamID	Position Aiding stream ID	-
4	F64	PosUpdateLat	Position update latitude	-
12	F64	PosUpdateLon	Position update longitude	-
20	F64	PosUpdateAlt	Position update altitude	m

Table 52:
Position aiding in the INS frame Message

ID: 43		Velocity aiding of the INS in navigation frame		
Version		0		
Description		Velocity aiding update of INS		
Header Timing		SDN		
Payload Size		29 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	S32	VelAidingStreamID	Velocity Aiding stream ID	-
4	U8	OdomStyleAiding	Aiding was odometry-style	-
5	F64	VelUpdateNorth	Velocity update north	m/s
13	F64	VelUpdateEast	Velocity update east	m/s
21	F64	VelUpdateDown	Velocity update down	m/s

Table 53:
Velocity aiding of the INS in navigation frame Message

ID: 44		Orientation aiding of the INS in dual-antenna frame		
Version		0		
Description		Orientation aiding update of INS		
Header Timing		SDN		
Payload Size		20 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	S32	AttAidingStreamID	Attitude Aiding stream ID	-
4	F64	AttPitchDiff	Orientation aiding pitch difference	rad
12	F64	AttHeadingDiff	Orientation aiding heading difference	rad

Table 54:
Orientation aiding of the INS in dual-antenna frame Message

ID: 45		Covariances of the latest position aiding update in navigation frame		
Version		0		
Description		Position aiding update covariances		
Header Timing		SDN		
Payload Size		52 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	S32	PosAidingStreamID	Position Aiding stream ID	-
4	F64	PosUpdateVarNorth	Position update variance north	-
12	F64	PosUpdateVarEast	Position update variance east	-
20	F64	PosUpdateVarDown	Position update variance down	-
28	F64	PosUpdateCovarEN	Position update covariance east-north	-
36	F64	PosUpdateCovarND	Position update covariance north-down	-
44	F64	PosUpdateCovarED	Position update covariance east-down	-

Table 55:
Covariances of the latest position aiding update in navigation frame Message

ID: 46		Covariances of the latest velocity aiding update in navigation frame		
Version		0		
Description		Velocity aiding update covariances		
Header Timing		SDN		
Payload Size		53 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	S32	VelAidingStreamID	Velocity Aiding stream ID	-
4	U8	OdomStyleAiding	Aiding was odometry-style	-
5	F64	VelUpdateVarNorth	Velocity update variance north	-
13	F64	VelUpdateVarEast	Velocity update variance east	-
21	F64	VelUpdateVarDown	Velocity update variance down	-
29	F64	VelUpdateCovarEN	Velocity update covariance east-north	-
37	F64	VelUpdateCovarND	Velocity update covariance north-down	-
45	F64	VelUpdateCovarED	Velocity update covariance east-down	-

Table 56:
Covariances of the latest velocity aiding update in navigation frame Message

ID: 47		Raw IMU measurements		
Version		0		
Description		Raw accelerations and angular rates in the IMU frame		
Header Timing		SDN		
Payload Size		48 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	RawAx	Raw X Acceleration	m/s^2
8	F64	RawAy	Raw Y Acceleration	m/s^2
16	F64	RawAz	Raw Z Acceleration	m/s^2
24	F64	RawWx	Raw X Angular rate	deg/s
32	F64	RawWy	Raw Y Angular rate	deg/s
40	F64	RawWz	Raw Z Angular rate	deg/s

Table 57:
Raw IMU measurements Message

ID: 48		OXTS vehicle frame linear jerks and angular accelerations		
Version		0		
Description		Linear jerks and angular accelerations in OXTS vehicle frame (XYZ) - the Kalman filter model has been applied to these signals		
Header Timing		SDN		
Payload Size		48 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	Jx	X Jerk	m/s ³
8	F64	Jy	Y Jerk	m/s ³
16	F64	Jz	Z Jerk	m/s ³
24	F64	Yx	X Angular Acceleration	deg/s ²
32	F64	Yy	Y Angular Acceleration	deg/s ²
40	F64	Yz	Z Angular Acceleration	deg/s ²

Table 58:
OXTS vehicle frame linear jerks and angular accelerations Message

ID: 49		OXTS navigation frame linear jerks and angular accelerations		
Version		0		
Description		Linear jerks and angular accelerations in OXTS navigation frame (NED) - the Kalman filter model has been applied to these signals		
Header Timing		SDN		
Payload Size		48 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	Jn	Jerk North	m/s ³
8	F64	Je	Jerk East	m/s ³
16	F64	Jd	Jerk down	m/s ³
24	F64	Yn	Angular acceleration North	deg/s ²
32	F64	Ye	Angular acceleration East	deg/s ²
40	F64	Yd	Angular acceleration down	deg/s ²

Table 59:
OXTS navigation frame linear jerks and angular accelerations Message

ID: 50		OXTS intermediate frame linear jerks and angular accelerations		
Version		0		
Description		Linear jerks and angular accelerations in OXTS intermediate frame (FLD) - the Kalman filter model has been applied to these signals		
Header Timing		SDN		
Payload Size		48 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	Jf	Jerk forward	m/s ³
8	F64	Jl	Jerk lateral	m/s ³
16	F64	Jd	Jerk down	m/s ³
24	F64	Yf	Angular acceleration forward	deg/s ²
32	F64	Yl	Angular acceleration lateral	deg/s ²
40	F64	Yd	Angular acceleration down	deg/s ²

Table 60:
OXTS intermediate frame linear jerks and angular accelerations Message

ID: 51		ISO-8855 Earth fixed linear jerks and angular accelerations		
Version		0		
Description		ISO-8855 Earth Fixed (ENU) inertial measurement - the Kalman filter model has been applied to these signals		
Header Timing		SDN		
Payload Size		48 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	IsoJnX	ISO e.f.s. east jerk	m/s ³
8	F64	IsoJnY	ISO e.f.s. north jerk	m/s ³
16	F64	IsoJnZ	ISO e.f.s. vertical jerk	m/s ³
24	F64	IsoYnX	ISO e.f.s. roll acceleration	deg/s ²
32	F64	IsoYnY	ISO e.f.s. pitch acceleration	deg/s ²
40	F64	IsoYnZ	ISO e.f.s. yaw acceleration	deg/s ²

Table 61:
ISO-8855 Earth fixed linear jerks and angular accelerations Message

ID: 52		ISO-8855 intermediate frame linear jerks and angular accelerations		
Version		0		
Description		ISO 8855 intermediate frame (FLD) linear jerks and angular accelerations - the Kalman filter model has been applied to these signals		
Header Timing		SDN		
Payload Size		48 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	IsoJhX	ISO i.s. longitudinal jerk	m/s ³
8	F64	IsoJhY	ISO i.s. lateral jerk	m/s ³
16	F64	IsoJhZ	ISO i.s. vertical jerk	m/s ³
24	F64	IsoYhX	ISO i.s. roll acceleration	deg/s ²
32	F64	IsoYhY	ISO i.s. pitch acceleration	deg/s ²
40	F64	IsoYhZ	ISO i.s. yaw acceleration	deg/s ²

Table 62:
ISO-8855 intermediate frame linear jerks and angular accelerations Message

ID: 53		ISO-8855 vehicle frame linear jerks and angular accelerations		
Version		0		
Description		ISO 8855 vehicle frame (XYZ) linear jerks and angular accelerations - the Kalman filter model has been applied to these signals		
Header Timing		SDN		
Payload Size		48 Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	F64	IsoJoX	ISO v.s. longitudinal jerk	m/s ³
8	F64	IsoJoY	ISO v.s. lateral jerk	m/s ³
16	F64	IsoJoZ	ISO v.s. vertical jerk	m/s ³
24	F64	IsoYoX	ISO v.s. roll acceleration	deg/s ²
32	F64	IsoYoY	ISO v.s. pitch acceleration	deg/s ²
40	F64	IsoYoZ	ISO v.s. yaw acceleration	deg/s ²

Table 63:
ISO-8855 vehicle frame linear jerks and angular accelerations Message

ID: 64510		Warnings		
Version		0		
Description		DBU Configuration warnings		
Header Timing		SDN		
Payload Size		Unknown Bytes		
Introduced In		UCOM Version 1		
Offset	Type	Signal ID	Name	Unit
0	U8	WarNo	Number of warnings	-
1	STR	WarMsg	Warning messages	-

Table 64:
Warnings Message

ID: 64511		Invalid mobile DBU		
Version		0		
Description		Invalid mobile DBU packet flag, containing the error messages encoded and separated by commas		
Header Timing		SDN		
Payload Size		Unknown Bytes		
Introduced In		UCOM Version 0		
Offset	Type	Signal ID	Name	Unit
0	U8	ErrNo	Number of errors	-
1	STR	ErrMsg	Error messages	-

Table 65:
Invalid mobile DBU Message

Sources And Their Signals

ID: TIME	Time stamp source		
Description	The resolved time stamps signals		
Signal ID	Signal Description	Native Type	Native Unit
Nano	The time according to the GNSS sources (previously GNSST)	S64	ns
SDNOFF	Offset from SDN time. First part of the signal is an Enum giving the offset source (byte 1). Second part is the offset in nano seconds (bytes 2-9). This signal cannot be added to any custom messages	EnS64	ns
UpTime	Time since power applied and system booted	S32	s
NanoFromInit	Time from initialisation	S64	ns

Table 66:
Time stamp source

ID: META	Meta source		
Description	Meta information from the INS		
Natural Output Type	OnChange		
Signal ID	Signal Description	Native Type	Native Unit
SerialNumber	The serial number of the INS	U32	-
UCOMVersion	The UCOM version of this UCOM output	U32	-
ErrNo	Number of invalid DBU error messages	U8	-
ErrMsg	The error messages describing why the DBU file is invalid	STR	-
WarNo	Number of DBU warning messages	U8	-
WarMsg	The warning messages describing any problems with the DBU	STR	-
DiskSpace	Remaining space on the internal disk	S32	KB
FileSize	Size of current logged raw data file	S32	KB
SupplyVolt	Supply voltage	F64	-
DevID	Software Version or Development ID that is running in the RT	STR	-
AdkDevID	Software Version or Advanced Development Kit Development ID that is running in the RT	STR	-

Table 67:
Meta source

ID: BNS_STS	BNS status source		
Description	Blended navigation system status		
Natural Output Type	OnChange		
Signal ID	Signal Description	Native Type	Native Unit
InsNavMode	Navigation status	U8	-
UmacStatus	Connection status based on position accuracy	U8	-
WiFiConnectionStatus	Connection status of Wi-Fi	U8	-
WiFiAccessPointStatus	Wi-Fi Access point status	U8	-
PTPStatus	Precision Time Protocol (PTP) Status	U8	-

Table 68:
BNS status source

ID: BNS_SDN	Strapdown navigator source		
Description	The signals produced by the strapdown navigator in the blended navigation system		
Natural Output Type	FrequencyBased		
Natural Output Frequency	IMU		
Signal ID	Signal Description	Native Type	Native Unit
Lat	Latitude	F64	deg
Lon	Longitude	F64	deg
Alt	Altitude	F64	m
Heading	Heading	F64	deg
Pitch	Pitch	F64	deg
Roll	Roll	F64	deg
Vx	Velocity in the x axis of the vehicle frame	F64	m/s
Vy	Velocity in the y axis of the vehicle frame	F64	m/s
Vz	Velocity in the z axis of the vehicle frame	F64	m/s
Ax	Acceleration in the x axis of the vehicle frame	F64	m/s^2
Ay	Acceleration along the y axis of the vehicle frame	F64	m/s^2
Az	Acceleration along the z axis of the vehicle frame	F64	m/s^2
Jx	Jerk in the x axis of the vehicle frame	F64	m/s^3
Jy	Jerk along the y axis of the vehicle frame	F64	m/s^3
Jz	Jerk along the z axis of the vehicle frame	F64	m/s^3
Wx	Angular rate along the x axis of the vehicle frame	F64	deg/s
Wy	Angular rate along the y axis of the vehicle frame	F64	deg/s
Wz	Angular rate along the z axis of the vehicle frame	F64	deg/s
Yx	Angular acceleration along the x axis of the vehicle frame	F64	deg/s^2

ID: BNS_SDN	Strapdown navigator source		
Yy	Angular acceleration in the y axis of the vehicle frame	F64	deg/s ²
Yz	Angular acceleration along the z axis of the vehicle frame	F64	deg/s ²
Vf	Velocity forward in the intermediate frame	F64	m/s
VI	Velocity lateral in the intermediate frame	F64	m/s
Af	Acceleration forward in the intermediate frame	F64	m/s ²
AI	Acceleration lateral in the intermediate frame	F64	m/s ²
Jf	Jerk forward in the intermediate frame	F64	m/s ³
JI	Jerk lateral in the intermediate frame	F64	m/s ³
Wf	Angular rate forward in the intermediate frame	F64	deg/s
WI	Angular rate lateral in the intermediate frame	F64	deg/s
Yf	Angular acceleration forward in the intermediate frame	F64	deg/s ²
YI	Angular acceleration lateral in the intermediate frame	F64	deg/s ²
Vn	Velocity North in the navigation frame	F64	m/s
Ve	Velocity East in the navigation frame	F64	m/s
Vd	Velocity down in the navigation frame and velocity down in the intermediate frame	F64	m/s
An	Acceleration North in the navigation frame	F64	m/s ²
Ae	Acceleration East in the navigation frame	F64	m/s ²
Ad	Acceleration down in the navigation frame and in the intermediate frame	F64	m/s ²
Jn	Jerk North in the navigation frame	F64	m/s ³
Je	Jerk East in the navigation frame	F64	m/s ³
Jd	Jerk in the down axis of the navigation frame and down in the intermediate frame	F64	m/s ³
Wn	Angular rate North in the navigation frame	F64	deg/s
We	Angular rate East in the navigation frame	F64	deg/s

ID: BNS_SDN	Strapdown navigator source		
Wd	Angular rate down in the navigation frame and the intermediate frame	F64	deg/s
Yn	Angular acceleration North in the navigation frame	F64	deg/s^2
Ye	Angular acceleration East in the navigation frame	F64	deg/s^2
Yd	Angular acceleration down in the navigation frame and in the intermediate frame	F64	deg/s^2
RawAx	Calibrated acceleration in the x axis of the IMU frame without Kalman Filter corrections applied	F64	m/s^2
RawAy	Calibrated acceleration in the y axis of the IMU frame without Kalman Filter corrections applied	F64	m/s^2
RawAz	Calibrated acceleration in the z axis of the IMU frame without Kalman Filter corrections applied	F64	m/s^2
RawWx	Calibrated angular rate along the x axis of the IMU frame without Kalman Filter corrections applied	F64	deg/s
RawWy	Calibrated angular rate along the y axis of the IMU frame without Kalman Filter corrections applied	F64	deg/s
RawWz	Calibrated angular rate along the z axis of the IMU frame without Kalman Filter corrections applied	F64	deg/s
Vu	OXTS NED frame vertical (up) velocity	F64	m/s
Undulation	Difference between ellipsoid and geoid	F64	m

Table 69:
Strapdown navigator source

ID: BNS_FLT	Filtered blended navigation source		
Description	Filtered signals from the blended navigation system		
Natural Output Type	FrequencyBased		
Natural Output Frequency	IMU		
Signal ID	Signal Description	Native Type	Native Unit
FiltAx	Filtered acceleration in the x axis of the vehicle frame	F64	m/s ²
FiltAy	Filtered acceleration along the y axis of the vehicle frame	F64	m/s ²
FiltAz	Filtered acceleration along the z axis of the vehicle frame	F64	m/s ²
FiltJx	Filtered jerk in the x axis of the vehicle frame	F64	m/s ³
FiltJy	Filtered jerk along the y axis of the vehicle frame	F64	m/s ³
FiltJz	Filtered jerk along the z axis of the vehicle frame	F64	m/s ³
FiltWx	Filtered angular rate along the x axis of the vehicle frame	F64	deg/s
FiltWy	Filtered angular rate along the y axis of the vehicle frame	F64	deg/s
FiltWz	Filtered angular rate along the z axis of the vehicle frame	F64	deg/s
FiltYx	Filtered angular acceleration along the x axis of the vehicle frame	F64	deg/s ²
FiltYy	Filtered angular acceleration in the y axis of the vehicle frame	F64	deg/s ²
FiltYz	Filtered angular acceleration along the z axis of the vehicle frame	F64	deg/s ²
FiltAf	Filtered acceleration forward in the intermediate frame	F64	m/s ²
FiltAl	Filtered acceleration lateral in the intermediate frame	F64	m/s ²
FiltJf	Filtered jerk forward in the intermediate frame	F64	m/s ³

ID: BNS_FLT	Filtered blended navigation source		
FiltJl	Filtered jerk lateral in the intermediate frame	F64	m/s ³
FiltWf	Filtered angular rate forward in the intermediate frame	F64	deg/s
FiltWl	Filtered angular rate lateral in the intermediate frame	F64	deg/s
FiltYf	Filtered angular acceleration forward in the intermediate frame	F64	deg/s ²
FiltYl	Filtered angular acceleration lateral in the intermediate frame	F64	deg/s ²
FiltAn	Filtered acceleration North in the navigation frame	F64	m/s ²
FiltAe	Filtered acceleration East in the navigation frame	F64	m/s ²
FiltAd	Filtered acceleration down in the navigation frame and in the intermediate frame	F64	m/s ²
FiltJn	Filtered jerk North in the navigation frame	F64	m/s ³
FiltJe	Filtered jerk East in the navigation frame	F64	m/s ³
FiltJd	Filtered jerk in the down axis of the navigation frame and the intermediate frame	F64	m/s ³
FiltWn	Filtered angular rate North in the navigation frame	F64	deg/s
FiltWe	Filtered angular rate East in the navigation frame	F64	deg/s
FiltWd	Filtered angular rate down in the navigation frame and the intermediate frame	F64	deg/s
FiltYn	Filtered angular acceleration North in the navigation frame	F64	deg/s ²
FiltYe	Filtered angular acceleration East in the navigation frame	F64	deg/s ²
FiltYd	Filtered angular acceleration down in the navigation frame and in the intermediate frame	F64	deg/s ²

Table 70:
Filtered blended navigation source

ID: BNS_DER	Derived signals source		
Description	Derived signals from the blended navigation system source		
Natural Output Type	FrequencyBased		
Natural Output Frequency	IMU		
Signal ID	Signal Description	Native Type	Native Unit
Heave	Real-time heave	F64	m
Curvature	Curvature of path	F64	1/m
Slip	Difference between orientation and actual movement direction	F64	deg
Track	Heading relative to the path	F64	deg
AngleGradient	Angular gradient	F64	deg
Dist2d	Horizontal distanced travelled including drift from INS	F64	m
Dist2dHold	Horizontal distanced travelled excluding drift from INS	F64	m
Dist3d	Horizontal distanced travelled including drift from INS	F64	m
Dist3dHold	Horizontal distanced travelled excluding drift from INS	F64	m
Speed2d	Horizontal, 2D speed	F64	m/s
Speed3d	All directional, 3D speed	F64	m/s
MeasPt1_Track	Measurement point 1 track angle	F64	deg
MeasPt2_Track	Measurement point 2 track angle	F64	deg
MeasPt3_Track	Measurement point 3 track angle	F64	deg
MeasPt4_Track	Measurement point 4 track angle	F64	deg
MeasPt5_Track	Measurement point 5 track angle	F64	deg
MeasPt6_Track	Measurement point 6 track angle	F64	deg
MeasPt7_Track	Measurement point 7 track angle	F64	deg
MeasPt8_Track	Measurement point 8 track angle	F64	deg
MeasPt1_Slip	Measurement point 1 slip angle	F64	deg

ID: BNS_DER	Derived signals source		
MeasPt2_Slip	Measurement point 2 slip angle	F64	deg
MeasPt3_Slip	Measurement point 3 slip angle	F64	deg
MeasPt4_Slip	Measurement point 4 slip angle	F64	deg
MeasPt5_Slip	Measurement point 5 slip angle	F64	deg
MeasPt6_Slip	Measurement point 6 slip angle	F64	deg
MeasPt7_Slip	Measurement point 7 slip angle	F64	deg
MeasPt8_Slip	Measurement point 8 slip angle	F64	deg
MeasPt1_Curvature	Measurement point 1 curvature	F64	1/m
MeasPt2_Curvature	Measurement point 2 curvature	F64	1/m
MeasPt3_Curvature	Measurement point 3 curvature	F64	1/m
MeasPt4_Curvature	Measurement point 4 curvature	F64	1/m
MeasPt5_Curvature	Measurement point 5 curvature	F64	1/m
MeasPt6_Curvature	Measurement point 6 curvature	F64	1/m
MeasPt7_Curvature	Measurement point 7 curvature	F64	1/m
MeasPt8_Curvature	Measurement point 8 curvature	F64	1/m
MeasPt1_Vf	Intermediate frame velocity forward at measurement point 1	F64	m/s
MeasPt1_Vl	Intermediate frame velocity lateral at measurement point 1	F64	m/s
MeasPt1_Vd	Intermediate frame velocity down at measurement point 1	F64	m/s
MeasPt1_Speed2d	Speed 2D at measurement point 1	F64	m/s
MeasPt1_An	NED frame acceleration north at measurement point 1	F64	m/s^2
MeasPt1_Ae	NED frame acceleration east at measurement point 1	F64	m/s^2
MeasPt1_Ad	NED and the FLD frame acceleration down at measurement point 1	F64	m/s^2
MeasPt1_Af	Intermediate frame acceleration forward at measurement point 1	F64	m/s^2
MeasPt1_Al	Intermediate frame acceleration lateral at measurement point 1	F64	m/s^2

ID: BNS_DER	Derived signals source		
MeasPt2_Vf	Intermediate frame velocity forward at measurement point 2	F64	m/s
MeasPt2_Vl	Intermediate frame velocity lateral at measurement point 2	F64	m/s
MeasPt2_Vd	Intermediate frame velocity down at measurement point 2	F64	m/s
MeasPt2_Speed2d	Speed 2D at measurement point 2	F64	m/s
MeasPt2_An	NED frame acceleration north at measurement point 2	F64	m/s ²
MeasPt2_Ae	NED frame acceleration east at measurement point 2	F64	m/s ²
MeasPt2_Ad	NED and the FLD frame acceleration down at measurement point 2	F64	m/s ²
MeasPt2_Af	Intermediate frame acceleration forward at measurement point 2	F64	m/s ²
MeasPt2_Al	Intermediate frame acceleration lateral at measurement point 2	F64	m/s ²
MeasPt3_Vf	Intermediate frame velocity forward at measurement point 3	F64	m/s
MeasPt3_Vl	Intermediate frame velocity lateral at measurement point 3	F64	m/s
MeasPt3_Vd	Intermediate frame velocity down at measurement point 3	F64	m/s
MeasPt3_Speed2d	Speed 2D at measurement point 3	F64	m/s
MeasPt3_An	NED frame acceleration north at measurement point 3	F64	m/s ²
MeasPt3_Ae	NED frame acceleration east at measurement point 3	F64	m/s ²
MeasPt3_Ad	NED and the FLD frame acceleration down at measurement point 3	F64	m/s ²
MeasPt3_Af	Intermediate frame acceleration forward at measurement point 3	F64	m/s ²
MeasPt3_Al	Intermediate frame acceleration lateral at measurement point 3	F64	m/s ²
MeasPt4_Vf	Intermediate frame velocity forward at measurement point 4	F64	m/s

ID: BNS_DER	Derived signals source		
MeasPt4_Vl	Intermediate frame velocity lateral at measurement point 4	F64	m/s
MeasPt4_Vd	Intermediate frame velocity down at measurement point 4	F64	m/s
MeasPt4_Speed2d	Speed 2D at measurement point 4	F64	m/s
MeasPt4_An	NED frame acceleration north at measurement point 4	F64	m/s^2
MeasPt4_Ae	NED frame acceleration east at measurement point 4	F64	m/s^2
MeasPt4_Ad	NED and the FLD frame acceleration down at measurement point 4	F64	m/s^2
MeasPt4_Af	Intermediate frame acceleration forward at measurement point 4	F64	m/s^2
MeasPt4_Al	Intermediate frame acceleration lateral at measurement point 4	F64	m/s^2
MeasPt5_Vf	Intermediate frame velocity forward at measurement point 5	F64	m/s
MeasPt5_Vl	Intermediate frame velocity lateral at measurement point 5	F64	m/s
MeasPt5_Vd	Intermediate frame velocity down at measurement point 5	F64	m/s
MeasPt5_Speed2d	Speed 2D at measurement point 5	F64	m/s
MeasPt5_An	NED frame acceleration north at measurement point 5	F64	m/s^2
MeasPt5_Ae	NED frame acceleration east at measurement point 5	F64	m/s^2
MeasPt5_Ad	NED and the FLD frame acceleration down at measurement point 5	F64	m/s^2
MeasPt5_Af	Intermediate frame acceleration forward at measurement point 5	F64	m/s^2
MeasPt5_Al	Intermediate frame acceleration lateral at measurement point 5	F64	m/s^2
MeasPt6_Vf	Intermediate frame velocity forward at measurement point 6	F64	m/s
MeasPt6_Vl	Intermediate frame velocity lateral at measurement point 6	F64	m/s

ID: BNS_DER	Derived signals source		
MeasPt6_Vd	Intermediate frame velocity down at measurement point 6	F64	m/s
MeasPt6_Speed2d	Speed 2D at measurement point 6	F64	m/s
MeasPt6_An	NED frame acceleration north at measurement point 6	F64	m/s^2
MeasPt6_Ae	NED frame acceleration east at measurement point 6	F64	m/s^2
MeasPt6_Ad	NED and the FLD frame acceleration down at measurement point 6	F64	m/s^2
MeasPt6_Af	Intermediate frame acceleration forward at measurement point 6	F64	m/s^2
MeasPt6_Al	Intermediate frame acceleration lateral at measurement point 6	F64	m/s^2
MeasPt7_Vf	Intermediate frame velocity forward at measurement point 7	F64	m/s
MeasPt7_Vl	Intermediate frame velocity lateral at measurement point 7	F64	m/s
MeasPt7_Vd	Intermediate frame velocity down at measurement point 7	F64	m/s
MeasPt7_Speed2d	Speed 2D at measurement point 7	F64	m/s
MeasPt7_An	NED frame acceleration north at measurement point 7	F64	m/s^2
MeasPt7_Ae	NED frame acceleration east at measurement point 7	F64	m/s^2
MeasPt7_Ad	NED and the FLD frame acceleration down at measurement point 7	F64	m/s^2
MeasPt7_Af	Intermediate frame acceleration forward at measurement point 7	F64	m/s^2
MeasPt7_Al	Intermediate frame acceleration lateral at measurement point 7	F64	m/s^2
MeasPt8_Vf	Intermediate frame velocity forward at measurement point 8	F64	m/s
MeasPt8_Vl	Intermediate frame velocity lateral at measurement point 8	F64	m/s
MeasPt8_Vd	Intermediate frame velocity down at measurement point 8	F64	m/s

ID: BNS_DER	Derived signals source		
MeasPt8_Speed2d	Speed 2D at measurement point 8	F64	m/s
MeasPt8_An	NED frame acceleration north at measurement point 8	F64	m/s^2
MeasPt8_Ae	NED frame acceleration east at measurement point 8	F64	m/s^2
MeasPt8_Ad	NED and the FLD frame acceleration down at measurement point 8	F64	m/s^2
MeasPt8_Af	Intermediate frame acceleration forward at measurement point 8	F64	m/s^2
MeasPt8_Al	Intermediate frame acceleration lateral at measurement point 8	F64	m/s^2
Speed2dHold	Speed in horizontal directions with hold	F64	m/s
Speed3dHold	Speed in all directions with hold	F64	m/s
Deprecated Signal ID	Recommended Alternative	Deprecated In	
LPHeave	Heave	UCOM Version 1	

Table 71:
Derived signals source

ID: BNS_ISO	ISO-8855 source		
Description	Signals from the blended navigation in the ISO 8855 frame		
Natural Output Type	FrequencyBased		
Natural Output Frequency	IMU		
Signal ID	Signal Description	Native Type	Native Unit
IsoYaw	ISO-8855 Yaw angle	F64	deg
IsoPitch	ISO-8855 Pitch angle	F64	deg
IsoRoll	ISO-8855 Roll angle	F64	deg
IsoVnX	ISO-8855 earth-fixed system east velocity	F64	m/s
IsoVnY	ISO-8855 earth-fixed system north velocity	F64	m/s
IsoVnZ	ISO-8855 earth-fixed system vertical velocity	F64	m/s
IsoVhX	ISO-8855 intermediate system longitudinal velocity	F64	m/s
IsoVhY	ISO-8855 intermediate system lateral velocity	F64	m/s
IsoVhZ	ISO-8855 intermediate system vertical velocity	F64	m/s
IsoVoX	ISO-8855 vehicle system longitudinal velocity	F64	m/s
IsoVoY	ISO-8855 vehicle system lateral velocity	F64	m/s
IsoVoZ	ISO-8855 vehicle system vertical velocity	F64	m/s
IsoAnX	ISO-8855 earth-fixed system east acceleration	F64	m/s ²
IsoAnY	ISO-8855 earth-fixed system north acceleration	F64	m/s ²
IsoAnZ	ISO-8855 earth-fixed system vertical acceleration	F64	m/s ²
IsoAhX	ISO-8855 intermediate system longitudinal acceleration	F64	m/s ²
IsoAhY	ISO-8855 intermediate system lateral acceleration	F64	m/s ²
IsoAhZ	ISO-8855 intermediate system vertical acceleration	F64	m/s ²

ID: BNS_ISO	ISO-8855 source		
IsoAoX	ISO-8855 vehicle system longitudinal acceleration	F64	m/s ²
IsoAoY	ISO-8855 vehicle system lateral acceleration	F64	m/s ²
IsoAoZ	ISO-8855 vehicle system vertical acceleration	F64	m/s ²
IsoJnX	ISO-8855 earth-fixed system east jerk	F64	m/s ³
IsoJnY	ISO-8855 earth-fixed system north jerk	F64	m/s ³
IsoJnZ	ISO-8855 earth-fixed system vertical jerk	F64	m/s ³
IsoJhX	ISO-8855 intermediate system longitudinal jerk	F64	m/s ³
IsoJhY	ISO-8855 intermediate system lateral jerk	F64	m/s ³
IsoJhZ	ISO-8855 intermediate system vertical jerk	F64	m/s ³
IsoJoX	ISO-8855 vehicle system longitudinal jerk	F64	m/s ³
IsoJoY	ISO-8855 vehicle system lateral jerk	F64	m/s ³
IsoJoZ	ISO-8855 vehicle system vertical jerk	F64	m/s ³
IsoWnX	ISO-8855 earth-fixed system roll velocity	F64	deg/s
IsoWnY	ISO-8855 earth-fixed system pitch velocity	F64	deg/s
IsoWnZ	ISO-8855 earth-fixed system yaw velocity	F64	deg/s
IsoWhX	ISO-8855 intermediate system roll velocity	F64	deg/s
IsoWhY	ISO-8855 intermediate system pitch velocity	F64	deg/s
IsoWhZ	ISO-8855 intermediate system yaw velocity	F64	deg/s
IsoWoX	ISO-8855 vehicle system roll velocity	F64	deg/s
IsoWoY	ISO-8855 vehicle system pitch velocity	F64	deg/s
IsoWoZ	ISO-8855 vehicle system yaw velocity	F64	deg/s
IsoYnX	ISO-8855 earth-fixed system roll acceleration	F64	deg/s ²
IsoYnY	ISO-8855 earth-fixed system pitch acceleration	F64	deg/s ²
IsoYnZ	ISO-8855 earth-fixed system yaw acceleration	F64	deg/s ²
IsoYhX	ISO-8855 intermediate system roll acceleration	F64	deg/s ²

ID: BNS_ISO	ISO-8855 source		
IsoYhY	ISO-8855 intermediate system pitch acceleration	F64	deg/s^2
IsoYhZ	ISO-8855 intermediate system yaw acceleration	F64	deg/s^2
IsoYoX	ISO-8855 vehicle system roll acceleration	F64	deg/s^2
IsoYoY	ISO-8855 vehicle system pitch acceleration	F64	deg/s^2
IsoYoZ	ISO-8855 vehicle system yaw acceleration	F64	deg/s^2

Table 72:
ISO-8855 source

ID: BNS_FLTISO	ISO-8855 filtered source		
Description	Filtered signals from the blended navigation in the ISO 8855 frame		
Natural Output Type	FrequencyBased		
Natural Output Frequency	IMU		
Signal ID	Signal Description	Native Type	Native Unit
FiltIsoAnX	Filtered ISO-8855 earth-fixed system east acceleration	F64	m/s ²
FiltIsoAnY	Filtered ISO-8855 earth-fixed system north acceleration	F64	m/s ²
FiltIsoAnZ	Filtered ISO-8855 earth-fixed system vertical acceleration	F64	m/s ²
FiltIsoAhX	Filtered ISO-8855 intermediate system longitudinal acceleration	F64	m/s ²
FiltIsoAhY	Filtered ISO-8855 intermediate system lateral acceleration	F64	m/s ²
FiltIsoAhZ	Filtered ISO-8855 intermediate system vertical acceleration	F64	m/s ²
FiltIsoAoX	Filtered ISO-8855 vehicle system longitudinal acceleration	F64	m/s ²
FiltIsoAoY	Filtered ISO-8855 vehicle system lateral acceleration	F64	m/s ²
FiltIsoAoZ	Filtered ISO-8855 vehicle system vertical acceleration	F64	m/s ²
FiltIsoJnX	Filtered ISO-8855 earth-fixed system east jerk	F64	m/s ³
FiltIsoJnY	Filtered ISO-8855 earth-fixed system north jerk	F64	m/s ³
FiltIsoJnZ	Filtered ISO-8855 earth-fixed system vertical jerk	F64	m/s ³
FiltIsoJhX	Filtered ISO-8855 intermediate system longitudinal jerk	F64	m/s ³
FiltIsoJhY	Filtered ISO-8855 intermediate system lateral jerk	F64	m/s ³
FiltIsoJhZ	Filtered ISO-8855 intermediate system vertical jerk	F64	m/s ³

ID: BNS_FLTISO	ISO-8855 filtered source		
FiltIsoJoX	Filtered ISO-8855 vehicle system longitudinal jerk	F64	m/s ³
FiltIsoJoY	Filtered ISO-8855 vehicle system lateral jerk	F64	m/s ³
FiltIsoJoZ	Filtered ISO-8855 vehicle system vertical jerk	F64	m/s ³
FiltWnX	Filtered ISO-8855 earth-fixed system roll velocity	F64	deg/s
FiltWnY	Filtered ISO-8855 earth-fixed system pitch velocity	F64	deg/s
FiltWnZ	Filtered ISO-8855 earth-fixed system yaw velocity	F64	deg/s
FiltWhX	Filtered ISO-8855 intermediate system roll velocity	F64	deg/s
FiltWhY	Filtered ISO-8855 intermediate system pitch velocity	F64	deg/s
FiltWhZ	Filtered ISO-8855 intermediate system yaw velocity	F64	deg/s
FiltWoX	Filtered ISO-8855 vehicle system roll velocity	F64	deg/s
FiltWoY	Filtered ISO-8855 vehicle system pitch velocity	F64	deg/s
FiltWoZ	Filtered ISO-8855 vehicle system yaw velocity	F64	deg/s
FiltIsoYnX	Filtered ISO-8855 earth-fixed system roll acceleration	F64	deg/s ²
FiltIsoYnY	Filtered ISO-8855 earth-fixed system pitch acceleration	F64	deg/s ²
FiltIsoYnZ	Filtered ISO-8855 earth-fixed system yaw acceleration	F64	deg/s ²
FiltIsoYhX	Filtered ISO-8855 intermediate system roll acceleration	F64	deg/s ²
FiltIsoYhY	Filtered ISO-8855 intermediate system pitch acceleration	F64	deg/s ²
FiltIsoYhZ	Filtered ISO-8855 intermediate system yaw acceleration	F64	deg/s ²
FiltIsoYoX	Filtered ISO-8855 vehicle system roll acceleration	F64	deg/s ²

ID: BNS_FLTISO	ISO-8855 filtered source		
FiltIsoYoY	Filtered ISO-8855 vehicle system pitch acceleration	F64	deg/s^2
FiltIsoYoZ	Filtered ISO-8855 vehicle system yaw acceleration	F64	deg/s^2

Table 73:
ISO-8855 filtered source

ID: BNS_ACC	Blended navigation system accuracies source		
Description	Accuracies of signals from the blended navigation system		
Natural Output Type	OnChange		
Signal ID	Signal Description	Native Type	Native Unit
NorthAcc	North accuracy	F64	m
EastAcc	East accuracy	F64	m
AltAcc	Down accuracy	F64	m
VnAcc	Velocity North accuracy	F64	m/s
VeAcc	Velocity East accuracy	F64	m/s
VdAcc	Velocity down accuracy	F64	m/s
VuAcc	Velocity up accuracy	F64	m/s
HeadingAcc	Heading accuracy	F64	deg
PitchAcc	Pitch accuracy	F64	deg
RollAcc	Roll accuracy	F64	deg

Table 74:
Blended navigation system accuracies source

ID: BNS_ACCISO	Blended navigation system ISO-8855 frame accuracies source		
Description	Accuracies of signals from the blended navigation system in the ISO-8855 frame		
Natural Output Type	OnChange		
Signal ID	Signal Description	Native Type	Native Unit
IsoYawAcc	Yaw accuracy	F64	deg
IsoPitchAcc	Pitch accuracy	F64	deg
IsoRollAcc	Roll accuracy	F64	deg
IsoVnXAcc	ENU East velocity accuracy	F64	m/s
IsoVnYAcc	ENU North velocity accuracy	F64	m/s
IsoVnZAcc	ENU vertical velocity accuracy	F64	m/s

Table 75:
Blended navigation system ISO-8855 frame accuracies source

ID: BNS_LOC	Local values source		
Description	Blended navigation signals in the local coordinate system		
Natural Output Type	FrequencyBased		
Natural Output Frequency	IMU		
Signal ID	Signal Description	Native Type	Native Unit
RefFrameY	Local Y position with respect to the local coordinate system	F64	m
RefFrameX	Local X position with respect to the local coordinate system	F64	m
RefFrameVelX	Local X velocity with respect to the local coordinate system	F64	m/s
RefFrameVelY	Local Y velocity with respect to the local coordinate system	F64	m/s
RefFrameTrack	Local track angle with respect to the local coordinate system	F64	deg
RefFrameYaw	Local yaw angle with respect to the local coordinate system	F64	deg
RefLat	Local Co-ordinates Origin Latitude	F64	deg
RefLon	Local Co-ordinates Origin Longitude	F64	deg
RefAlt	Local Co-ordinates Origin Altitude	F64	m
RefHeading	Local Co-ordinates Origin Heading	F64	deg
RefFrameNorthing	Local co-ordinates northing	F64	m
RefFrameEasting	Local co-ordinates easting	F64	m

Table 76:
Local values source

ID: GNSS_STS	GNSS status source		
Description	GNSS status including navigation status and GNSS tracking		
Natural Output Type	FrequencyBased		
Natural Output Frequency	GNSSRaw		
Signal ID	Signal Description	Native Type	Native Unit
GnssPosMode	Position mode	U8	-
GnssVelMode	Velocity mode	U8	-
GnssAttMode	Attitude mode	U8	-
GnssDiffAge	Age of current differential corrections	F64	s
PriSatTrac	Satellites tracked by the primary GNSS card	U8	-
SecSatTrac	Satellites tracked by the secondary GNSS card	U8	-
GnssPosNumSats	Satellites used in the position solution	U8	-
HeaSatTrac	Satellites tracked for attitude	U8	-
GnssAttNumSats	Satellites used for attitude solution	U8	-
GnssDiffNumStations	Differential GNSS number of stations	S32	-
GnssDiffNumGpsL1	Raw differential corrections GPS L1	U8	-
GnssDiffNumGpsL2	Raw differential corrections GPS L2	U8	-
GnssDiffNumGlonassL1	Raw differential corrections GLONASS L1	U8	-
GnssDiffNumGlonassL2	Raw differential corrections GLONASS L2	U8	-
GnssDiffNumGalileoE1	Number of Galileo E1 measurements received in differential corrections	U8	-
GnssDiffNumGalileoE5b	Number of Galileo E5b measurements received in differential corrections	U8	-
GnssDiffNumBeiDou2B1	Number of BeiDou Gen 2 B1 measurements received in differential corrections	U8	-
GnssDiffNumBeiDou2B2	Number of BeiDou Gen 2 B2 measurements received in differential corrections	U8	-
GnssDiffNumBeiDou3B1	Number of BeiDou Gen 3 B1 measurements received in differential corrections	U8	-

ID: GNSS_STS	GNSS status source		
HeadNumSats	Total number of Satellites available for attitude	U8	-
HeadNumSatsGps	Total number of GPS Satellites available for attitude	U8	-
HeadNumSatsGlonass	Total number of Glonass Satellites available for attitude	U8	-
HeadNumSatsGalileo	Total number of Galileo Satellites available for attitude	U8	-
HeadNumSatsBeiDou2	Total number of BeiDou Gen 2 Satellites available for attitude	U8	-
HeadNumSatsBeiDou3	Total number of BeiDou Gen 3 Satellites available for attitude	U8	-
HeadNumSatsUsed	Total number of Satellites used for attitude	U8	-
HeadNumSatsUsedGps	Total number of GPS Satellites used for attitude	U8	-
HeadNumSatsUsedGlonass	Total number of Glonass Satellites used for attitude	U8	-
HeadNumSatsUsedGalileo	Total number of Galileo Satellites used for attitude	U8	-
HeadNumSatsUsedBeiDou2	Total number of BeiDou Gen 2 Satellites used for attitude	U8	-
HeadNumSatsUsedBeiDou3	Total number of BeiDou Gen 3 Satellites used for attitude	U8	-
HeadNumGpsL1	Total number of GPS L1 measurements available for attitude	U8	-
HeadNumGpsL2	Total number of GPS L2 measurements available for attitude	U8	-
HeadNumGlonassL1	Total number of GLONASS L1 measurements available for attitude	U8	-
HeadNumGlonassL2	Total number of GLONASS L2 measurements available for attitude	U8	-
HeadNumGalileoE1	Total number of GALILEO E1 measurements available for attitude	U8	-
HeadNumGalileoE5b	Total number of GALILEO E5b measurements available for attitude	U8	-

ID: GNSS_STS	GNSS status source		
HeadNumBeiDou2B1	Total number of BEIDOU GEN 2 B1 measurements available for attitude	U8	-
HeadNumBeiDou2B2	Total number of BEIDOU GEN 2 B2 measurements available for attitude	U8	-
HeadNumBeiDou3B1	Total number of BEIDOU GEN 3 B1 measurements available for attitude	U8	-
HeadNumUsedGpsL1	Number of L1 GPS measurements used for attitude	U8	-
HeadNumUsedGpsL2	Number of L2 GPS measurements used for attitude	U8	-
HeadNumUsedGlonassL1	Number of L1 GLONASS measurements used for attitude	U8	-
HeadNumUsedGlonassL2	Number of L2 GLONASS measurements used for attitude	U8	-
HeadNumUsedGalileoE1	Number of E1 GALILEO measurements used for attitude	U8	-
HeadNumUsedGalileoE5b	Number of E5b GALILEO measurements used for attitude	U8	-
HeadNumUsedBeiDou2B1	Total number of BEIDOU GEN 2 B1 measurements used for attitude	U8	-
HeadNumUsedBeiDou2B2	Total number of BEIDOU GEN 2 B2 measurements used for attitude	U8	-
HeadNumUsedBeiDou3B1	Total number of BEIDOU GEN 3 B1 measurements used for attitude	U8	-
GnssVelNumSatsUsed	Number of satellites used in velocity solution	U8	-
GnssPosNumSatsUsed	Number of satellites used in position solution	U8	-
GnssPosNumSatsUsedGps	Number of Gps satellites used in position solution	U8	-
GnssPosNumSatsUsedGlonass	Number of Glonass satellites used in position solution	U8	-
GnssPosNumSatsUsedGalileo	Number of Galileo satellites used in position solution	U8	-
GnssPosNumSatsUsedBeiDou2	Number of BeiDou Gen 2 satellites used in position solution	U8	-
GnssPosNumSatsUsedBeiDou3	Number of BeiDou Gen 3 satellites used in position solution	U8	-

ID: GNSS_STS	GNSS status source		
GnssPosNumGpsL1	Number of Gps L1 signals available in position solution	U8	-
GnssPosNumGpsL2	Number of Gps L2 signals available in position solution	U8	-
GnssPosNumGlonassL1	Number of Glonass L1 signals available in position solution	U8	-
GnssPosNumGlonassL2	Number of Glonass L2 signals available in position solution	U8	-
GnssPosNumGalileoE1	Number of Galileo E1 signals available in position solution	U8	-
GnssPosNumGalileoE5b	Number of Galileo E5b signals available in position solution	U8	-
GnssPosNumBeiDou2B1	Number of BeiDou Gen 2 B1 signals available in position solution	U8	-
GnssPosNumBeiDou2B2	Number of BeiDou Gen 2 B2 signals available in position solution	U8	-
GnssPosNumBeiDou3B1	Number of BeiDou Gen 3 B1 signals available in position solution	U8	-
GnssPosNumUsedGpsL1	Number of Gps L1 signals used in position solution	U8	-
GnssPosNumUsedGpsL2	Number of Gps L2 signals used in position solution	U8	-
GnssPosNumUsedGlonassL1	Number of Glonass L1 signals used in position solution	U8	-
GnssPosNumUsedGlonassL2	Number of Glonass L2 signals used in position solution	U8	-
GnssPosNumUsedGalileoE1	Number of Galileo E1 signals used in position solution	U8	-
GnssPosNumUsedGalileoE5b	Number of Galileo E5b signals used in position solution	U8	-
GnssPosNumUsedBeiDou2B1	Number of BeiDou Gen 2 B1 signals used in position solution	U8	-
GnssPosNumUsedBeiDou2B2	Number of BeiDou Gen 2 B2 signals used in position solution	U8	-
GnssPosNumUsedBeiDou3B1	Number of BeiDou Gen 3 B1 signals used in position solution	U8	-

ID: GNSS_STS	GNSS status source		
GnssDiffNtripStatus	NTRIP state machine status	U8	-
PPSStatus	PPS Status	U8	-

Table 77:
GNSS status source

ID: GNSS_PGC	Primary GNSS card source		
Description	Signals from the primary GNSS card		
Natural Output Type	FrequencyBased		
Natural Output Frequency	GNSSRaw		
Signal ID	Signal Description	Native Type	Native Unit
PGCApproxLat	Primary GNSS Card approximate latitude	F64	-
PGCApproxLon	Primary GNSS Card approximate longitude	F64	deg
PGCApproxAlt	Primary GNSS Card approximate altitude	F64	m
PGCVd	Primary GNSS Card approximate velocity down in the navigation frame	F64	m/s
PGCVe	Primary GNSS Card approximate velocity East in the navigation frame	F64	m/s
PGCVn	Primary GNSS Card approximate velocity North in the navigation frame	F64	m/s
GnssInt1_AntPower	GNSS Antenna Power Status	S8	-
GnssInt1_NumSats	Number of Satellites Tracked	U8	-
GnssInt1_Chars	Characters received from GNSS1 by the navigation computer	U32	-
GnssInt1_CharsSkipped	Characters received from GNSS1 by the navigation computer but not understood by the decoder	U32	-
GnssInt1_Pkts	Packets received from GNSS1 by the navigation computer	U32	-
GnssInt1_OldPkts	Packets received from GNSS1 by the navigation computer that could not be used to update the Kalman filter (e.g too old)	U32	-
GnssInt1_NumGpsL1	Internal1 receiver GPS L1 number of raw measurements received	U8	-
GnssInt1_NumGpsL2	Internal1 receiver GPS L2 number of raw measurements received	U8	-
GnssInt1_NumGlonassL1	Internal1 receiver GLONASS L1 number of raw measurements received	U8	-
GnssInt1_NumGlonassL2	Internal1 receiver GLONASS L2 number of raw measurements received	U8	-

ID: GNSS_PGC	Primary GNSS card source		
GnssInt1_NumGalileoE1	Internal1 receiver Galileo E1 number of raw measurements received	U8	-
GnssInt1_NumGalileoE5b	Internal1 receiver Galileo E5b number of raw measurements received	U8	-
GnssInt1_NumBeiDou2B1	Internal1 receiver BeiDou Gen 2 B1 number of raw measurements received	U8	-
GnssInt1_NumBeiDou2B2	Internal1 receiver BeiDou Gen 2 B2 number of raw measurements received	U8	-
GnssInt1_NumBeiDou3B1	Internal1 receiver BeiDou Gen 3 B1 number of raw measurements received	U8	-
Deprecated Signal ID	Recommended Alternative	Deprecated In	
PGCHHeading	Heading	UCOM Version 1	
PGCPitch	Pitch	UCOM Version 1	
PGCRoll	Roll	UCOM Version 1	

Table 78:
Primary GNSS card source

ID: GNSS_SGC	Secondary GNSS card source		
Description	Signals from the secondary GNSS card		
Natural Output Type	FrequencyBased		
Natural Output Frequency	GNSSRaw		
Signal ID	Signal Description	Native Type	Native Unit
SGCApproxLat	Secondary GNSS Card approximate latitude	F64	deg
SGCApproxLon	Secondary GNSS Card approximate longitude	F64	deg
SGCApproxAlt	Secondary GNSS Card approximate altitude	F64	m
GnssInt2_AntPower	GNSS Antenna Power Status	S8	-
GnssInt2_NumSats	Number of Satellites Tracked	U8	-
GnssInt2_Chars	Characters received from GNSS2 by the navigation computer	U32	-
GnssInt2_CharsSkipped	Characters received from GNSS2 by the navigation computer but not understood by the decoder	U32	-
GnssInt2_Pkts	Packets received from GNSS2 by the navigation computer	U32	-
GnssInt2_OldPkts	Packets received from GNSS2 by the navigation computer that could not be used to update the Kalman filter (e.g too old)	U32	-
GnssInt2_NumGpsL1	Internal2 receiver GPS L1 number of raw measurements received	U8	-
GnssInt2_NumGlonassL1	Internal2 receiver GLONASS L1 number of raw measurements received	U8	-
GnssInt2_NumGpsL2	Internal2 receiver GPS L2 number of raw measurements received	U8	-
GnssInt2_NumGlonassL2	Internal2 receiver GLONASS L2 number of raw measurements received	U8	-
GnssInt2_NumGalileoE1	Internal2 receiver Galileo E1 number of raw measurements received	U8	-
GnssInt2_NumGalileoE5b	Internal2 receiver Galileo E5b number of raw measurements received	U8	-
GnssInt2_NumBeiDou2B1	Internal2 receiver BeiDou Gen 2 B1 number of raw measurements received	U8	-

ID: GNSS_SGC	Secondary GNSS card source		
GnssInt2_NumBeiDou2B2	Internal2 receiver BeiDou Gen 2 B2 number of raw measurements received	U8	-
GnssInt2_NumBeiDou3B1	Internal2 receiver BeiDou Gen 3 B1 number of raw measurements received	U8	-

Table 79:
Secondary GNSS card source

ID: AID_INNOV	Innovations		
Description	Innovations		
Natural Output Type	OnChange		
Signal ID	Signal Description	Native Type	Native Unit
InnHeadingH	Heading Lock Innovation	F64	-
InnPosX	Position X Innovation	F64	-
InnPosY	Position Y Innovation	F64	-
InnPosZ	Position Z Innovation	F64	-
InnVelX	Velocity X Innovation	F64	-
InnVelY	Velocity Y Innovation	F64	-
InnVelZ	Velocity Z Innovation	F64	-
InnHeading	Orientation Heading Innovation	F64	-
InnPitch	Orientation Pitch Innovation	F64	-
InnZeroVelX	Zero Velocity X Innovation	F64	-
InnZeroVelY	Zero Velocity Y Innovation	F64	-
InnZeroVelZ	Zero Velocity Z Innovation	F64	-
InnNoSlipH	Lateral Advanced Slip Innovation	F64	-
InnNoVSlipP	Vertical Advanced Slip Innovation	F64	-
InnWSpeed	Wheel Speed Innovation	F64	-
PosAidingStreamID	ID of the stream providing position aiding to the INS	S32	-
VelAidingStreamID	ID of the stream providing velocity aiding to the INS	S32	-
AttAidingStreamID	ID of the stream providing attitude aiding to the INS	S32	-
PosUpdateLat	WGS84 Latitude of the IMU implied by the latest position update	F64	-
PosUpdateLon	WGS84 Longitude of the IMU implied by the latest position update	F64	-

ID: AID_INNOV	Innovations		
PosUpdateAlt	WGS84 Altitude of the IMU implied by the latest position update	F64	m
PosUpdateVarNorth	Variance applied to the north component of the latest position update. This includes components from the update itself and vehicle vibration	F64	-
PosUpdateVarEast	Variance applied to the east component of the latest position update. This includes components from the update itself and vehicle vibration	F64	-
PosUpdateVarDown	Variance applied to the down component of the latest position update. This includes components from the update itself and vehicle vibration	F64	-
PosUpdateCovarEN	East-north component of the covariance matrix of the latest position update. This includes components from the update itself and vehicle vibration	F64	-
PosUpdateCovarND	North-down component of the covariance matrix of the latest position update. This includes components from the update itself and vehicle vibration	F64	-
PosUpdateCovarED	East-down component of the covariance matrix of the latest position update. This includes components from the update itself and vehicle vibration	F64	-
VelUpdateNorth	Velocity north of the INS implied by the latest velocity update	F64	m/s
VelUpdateEast	Velocity east of the INS implied by the latest velocity update	F64	m/s
VelUpdateDown	Velocity down of the INS implied by the latest velocity update	F64	m/s
OdomStyleAiding	Was the velocity update provided in the odometry frame (attached to the vehicle), as opposed to the navigation frame?	U8	-
VelUpdateVarNorth	Variance applied to the north component of the latest velocity update. This includes components from the update itself and vehicle vibration	F64	-
VelUpdateVarEast	Variance applied to the east component of the latest velocity update. This includes	F64	-

ID: AID_INNOV	Innovations		
	components from the update itself and vehicle vibration		
VelUpdateVarDown	Variance applied to the down component of the latest velocity update. This includes components from the update itself and vehicle vibration	F64	-
VelUpdateCovarEN	East-north component of the covariance matrix of the latest velocity update. This includes components from the update itself and vehicle vibration	F64	-
VelUpdateCovarND	North-down component of the covariance matrix of the latest velocity update. This includes components from the update itself and vehicle vibration	F64	-
VelUpdateCovarED	East-down component of the covariance matrix of the latest velocity update. This includes components from the update itself and vehicle vibration	F64	-
AttHeadingDiff	Difference between orientation aiding heading and Kalman filter prediction	F64	rad
AttPitchDiff	Difference between orientation aiding pitch and Kalman filter prediction	F64	rad

Table 80:
Innovations

ID: INS_CFG	INS Configuration source		
Description	The source with all the configuration information about the inertial navigation system		
Natural Output Type	OnChange		
Signal ID	Signal Description	Native Type	Native Unit
ImuRate	Maximum output rate of the IMU	F64	-
OutputRegularRate	Current output rate of the IMU	F64	-
OptionUdp	Configuration of Ethernet UDP1 output	S8	-
OptionSer1	Configuration of Serial 1 output (NCOM/NMEA, etc)	S8	-
OptionSer2	Configuration of Serial 2 output (NCOM/NMEA, etc)	S8	-
OptionSer3	Configuration of Serial 3 output (NCOM/NMEA, etc)	S8	-
OptionSer1Baud	Serial 1 baud rate	S32	-
OptionSer1DataBits	Serial 1 data bits	S32	-
OptionSer1StopBits	Serial 1 stop bits	S32	-
OptionSer1Parity	Serial 1 parity	S32	-
OptionSer2Baud	Serial 2 baud rate	S32	-
OptionSer2DataBits	Serial 2 data bits	S32	-
OptionSer2StopBits	Serial 2 stop bits	S32	-
OptionSer2Parity	Serial 2 parity	S32	-
OptionSer3Baud	Serial 3 baud rate	S32	-
OptionSer3DataBits	Serial 3 data bits	S32	-
OptionSer3StopBits	Serial 3 stop bits	S32	-
OptionSer3Parity	Serial 3 parity	S32	-
OptionCanBaud	CAN bus baud rate	S32	-
TimeSource	Source of time used by the Inertial Navigation System	S32	-
RemoteLeverArmX	Output displacement lever-arm X	F64	m

ID: INS_CFG	INS Configuration source		
RemoteLeverArmY	Output displacement lever-arm Y	F64	m
RemoteLeverArmZ	Output displacement lever-arm Z	F64	m
ZeroVelLeverArmX	Distance to the Zero Velocity Point in X direction	F64	m
ZeroVelLeverArmY	Distance to the Zero Velocity Point in Y direction	F64	m
ZeroVelLeverArmZ	Distance to the Zero Velocity Point in Z direction	F64	m
ZeroVelLeverArmXAcc	Accuracy of Distance to the Zero Velocity Point in X direction	F64	m
ZeroVelLeverArmYAcc	Accuracy of Distance to the Zero Velocity Point in Y direction	F64	m
ZeroVelLeverArmZAcc	Accuracy of Distance to the Zero Velocity Point in Z direction	F64	m
NoSlipLeverArmX	Distance to the Vertical Advanced Slip Point in X direction	F64	m
NoSlipLeverArmY	Distance to the Vertical Advanced Slip Point in Y direction	F64	m
NoSlipLeverArmZ	Distance to the Vertical Advanced Slip Point in Z direction	F64	m
NoSlipLeverArmXAcc	Accuracy of Distance to the Vertical Advanced Slip Point in X direction	F64	m
NoSlipLeverArmYAcc	Accuracy of Distance to the Vertical Advanced Slip Point in Y direction	F64	m
NoSlipLeverArmZAcc	Accuracy of Distance to the Vertical Advanced Slip Point in Z direction	F64	m
NoVSlipLeverArmX	Distance to the Lateral Advanced Slip Point in X direction	F64	m
NoVSlipLeverArmY	Distance to the Lateral Advanced Slip Point in Y direction	F64	m
NoVSlipLeverArmZ	Distance to the Lateral Advanced Slip Point in Z direction	F64	m
NoVSlipLeverArmXAcc	Accuracy of Distance to the Lateral Advanced Slip Point in X direction	F64	m
NoVSlipLeverArmYAcc	Accuracy of Distance to the Lateral Advanced Slip Point in Y direction	F64	m

ID: INS_CFG	INS Configuration source		
NoVSlipLeverArmZAcc	Accuracy of Distance to the Lateral Advanced Slip Point in Z direction	F64	m
MeasPt1_PointXv	Distance to the Additional Slip Point in X direction	F64	m
MeasPt2_PointXv	Distance to the Additional Slip Point in X direction	F64	m
MeasPt3_PointXv	Distance to the Additional Slip Point in X direction	F64	m
MeasPt4_PointXv	Distance to the Additional Slip Point in X direction	F64	m
MeasPt5_PointXv	Distance to the Additional Slip Point in X direction	F64	m
MeasPt6_PointXv	Distance to the Additional Slip Point in X direction	F64	m
MeasPt7_PointXv	Distance to the Additional Slip Point in X direction	F64	m
MeasPt8_PointXv	Distance to the Additional Slip Point in X direction	F64	m
MeasPt1_PointYv	Distance to the Additional Slip Point in Y direction	F64	m
MeasPt2_PointYv	Distance to the Additional Slip Point in Y direction	F64	m
MeasPt3_PointYv	Distance to the Additional Slip Point in Y direction	F64	m
MeasPt4_PointYv	Distance to the Additional Slip Point in Y direction	F64	m
MeasPt5_PointYv	Distance to the Additional Slip Point in Y direction	F64	m
MeasPt6_PointYv	Distance to the Additional Slip Point in Y direction	F64	m
MeasPt7_PointYv	Distance to the Additional Slip Point in Y direction	F64	m
MeasPt8_PointYv	Distance to the Additional Slip Point in Y direction	F64	m
MeasPt1_PointZv	Distance to the Additional Slip Point in Z direction	F64	m

ID: INS_CFG	INS Configuration source		
MeasPt2_PointZv	Distance to the Additional Slip Point in Z direction	F64	m
MeasPt3_PointZv	Distance to the Additional Slip Point in Z direction	F64	m
MeasPt4_PointZv	Distance to the Additional Slip Point in Z direction	F64	m
MeasPt5_PointZv	Distance to the Additional Slip Point in Z direction	F64	m
MeasPt6_PointZv	Distance to the Additional Slip Point in Z direction	F64	m
MeasPt7_PointZv	Distance to the Additional Slip Point in Z direction	F64	m
MeasPt8_PointZv	Distance to the Additional Slip Point in Z direction	F64	m
WiFiIPAddress	Configured IP address for Wi-Fi	U8	-
WiFiRegion	Configured WiFi region	U8	-
NMEAOutputEnabled	Is the device configured to output NMEA?	U8	-
PTPEpoch	PTP epoch type	S32	-
LidarIPAddress	Least significant digits of configured LiDAR device	U8	-

Table 81:
INS Configuration source

ID: BNS_CFG	BNS Configuration source		
Description	The source with all the configuration information about the blended navigation system		
Natural Output Type	OnChange		
Signal ID	Signal Description	Native Type	Native Unit
Generator	Blending processing method	S32	-
OptionLevel	Configuration for level or not level	S8	-
OptionVibration	Configuration of vibration	S8	-
OptionHeading	Configuration of static initialisation	S8	-
OptionInitSpeed	Configuration of the initialisation speed	F64	m/s
OptionTopSpeed	Top speed option	F64	m/s
LinAccFiltFreq	Parameters for the (linear) acceleration filter	F64	-
LinAccFiltZeta	Parameters for the (linear) acceleration filter	F64	-
AngAccFiltFreq	Parameters for the angular acceleration filter	F64	-
AngAccFiltZeta	Parameters for the angular acceleration filter	F64	-
OptionSZVDelay	Garage mode zero velocity delay	F64	s
OptionSZVPeriod	Garage mode zero velocity period	F64	s
OptionNSDelay	Lateral advanced slip delay before update	F64	s
OptionNSPeriod	Lateral advanced slip period of update	F64	s
OptionNSAngleStd	Lateral advanced slip angle weight	F64	deg
OptionNSHAccel	Lateral advanced slip maximum horizontal acceleration	F64	m/s^2
OptionNSVAccel	Lateral advanced slip maximum vertical acceleration	F64	m/s^2
OptionNSSpeed	Lateral advanced slip minimum speed	F64	m/s
OptionNSRadius	Lateral advanced slip minimum radius	F64	m
OptionNVSDelay	Vertical advanced slip delay before update	F64	s
OptionNVSPeriod	Vertical advanced slip period of update	F64	s
OptionNVSAngleStd	Vertical advanced slip angle weight	F64	deg

ID: BNS_CFG	BNS Configuration source		
OptionNVSVAccel	Vertical advanced slip maximum vertical acceleration	F64	m/s^2
OptionNVSSpeed	Vertical advanced slip minimum speed	F64	m/s
OptionNVSRadius	Vertical advanced slip minimum radius	F64	m
OptionHLDelay	Heading lock delay before update	F64	s
OptionHLPeriod	Heading lock period of update	F64	s
OptionHLAngleStd	Heading lock weight	F64	deg
OptionStatDelay	Time elapsed before assumed stationary	F64	s
OptionStatSpeed	Stationary speed threshold	F64	m/s
HeaveDCCoupling	Is Heave DC coupled?	F64	-
OptionHeave	Heave coupling mode	S8	-
DatumEllipsoid	Ellipsoid	S8	-
DatumFrame	Earth frame associated with datum	S8	-
OutputDatum	Configured output co-ordinate datum	S8	-
LinJerkFiltType	Linear jerk filter type	U8	-
LinJerkMovAvgLen	Linear jerk moving average time interval	F32	s
LinJerkFiltFreq	Linear jerk filter cut-off frequency	F32	Hz
LinJerkFiltOrder	Linear jerk filter order	U8	-
Imu2VehHeading	Heading of the vehicle in the RT co-ordinate frame	F64	deg
Imu2VehPitch	Pitch of the vehicle in the RT co-ordinate frame	F64	deg
Imu2VehRoll	Roll of the vehicle in the RT co-ordinate frame	F64	deg
Veh2OutHeading	Vehicle to output rotation heading	F64	deg
Veh2OutPitch	Vehicle to output rotation pitch	F64	deg
Veh2OutRoll	Vehicle to output rotation roll	F64	deg
Surf2OutHeading	Heading to surface	F64	deg
Surf2OutPitch	Pitch to surface	F64	deg
Surf2OutRoll	Roll to surface	F64	deg
Ned2SurfHeading	Surface Angle Heading Rotation	F64	deg

ID: BNS_CFG	BNS Configuration source		
Ned2SurfPitch	Surface Angle Pitch Rotation	F64	deg
Ned2SurfRoll	Surface Angle Roll Rotation	F64	deg

Table 82:
BNS Configuration source

ID: GNSS_CFG	GNSS Configuration source		
Description	The source with all the configuration information about the GNSS engine		
Natural Output Type	OnChange		
Signal ID	Signal Description	Native Type	Native Unit
GnssDiffFormat	Format of the current differential corrections (RTCA, RTCM, etc)	U8	-
GnssDiffType	Differential GNSS type	U8	-
BaseLineLength	Distance between the GNSS Antennas	F64	m
BaseLineLengthAcc	Accuracy of Distance between the GNSS Antennas	F64	m

Table 83:
GNSS Configuration source

ID: BNS_OPT	Blended navigation system optimised parameters source		
Description	Optimised internal parameters of the blended navigation system		
Natural Output Type	OnChange		
Signal ID	Signal Description	Native Type	Native Unit
GAPx	Distance to Primary GNSS Antenna in X direction	F64	m
GAPy	Distance to Primary GNSS Antenna in Y direction	F64	m
GAPz	Distance to Primary GNSS Antenna in Z direction	F64	m
AtH	Heading Orientation of the GNSS Antenna	F64	deg
AtP	Pitch Orientation of the GNSS Antenna	F64	deg
HeadingMisAlign	Heading Misalignment Angle	F64	deg
PitchMisAlign	Pitch Misalignment Angle	F64	deg

Table 84:
Blended navigation system optimised parameters source

ID: BNS_ACCOPT	Blended navigation system optimised parameter accuracies source		
Description	Optimised internal parameter accuracies of the blended navigation system		
Natural Output Type	OnChange		
Signal ID	Signal Description	Native Type	Native Unit
GAPxAcc	Accuracy of Distance to Primary GNSS Antenna in X direction	F64	m
GAPyAcc	Accuracy of Distance to Primary GNSS Antenna in Y direction	F64	m
GAPzAcc	Accuracy of Distance to Primary GNSS Antenna in Z direction	F64	m
AtHAcc	Accuracy of Heading Orientation of the GNSS Antenna	F64	deg
AtPAcc	Accuracy of Pitch Orientation of the GNSS Antenna	F64	deg
HeadingMisAlignAcc	Accuracy of Heading Misalignment Angle	F64	deg
PitchMisAlignAcc	Accuracy of Pitch Misalignment Angle	F64	deg

Table 85:
Blended navigation system optimised parameter accuracies source

ID: BNS_TRG	Blended navigation trigger source		
Description	The trigger signals from the blended navigation system		
Natural Output Type	OnTrigger		
Signal ID	Signal Description	Native Type	Native Unit
Trig1FallUpdateCount	Update count of trigger 1 falling edge	S32	-
TriggerDistanceFalling	Distance with hold since last trigger	F64	m
TriggerDistanceRising	Distance with hold since last trigger	F64	m
Trig1RiseUpdateCount	Update count of trigger 1 rising edge	S32	-
Trig2FallUpdateCount	Update count of trigger 2 falling edge	S32	-
Trig2RiseUpdateCount	Update count of trigger 2 rising edge	S32	-
Trig1TranLevel	Level after trigger 1 transition	S8	-
Trig2TranLevel	Level after trigger 2 transition	S8	-
Trig1TranUpdateCount	Update count of trigger 1 transition	U32	-
Trig2TranUpdateCount	Update count of trigger 2 transition	U32	-

Table 86:
Blended navigation trigger source

ID: BNS_STATS	Blended navigation statistics source		
Description	The statistics from the blended navigation system		
Natural Output Type	OnChange		
Signal ID	Signal Description	Native Type	Native Unit
CanTxNum	Number of CAN messages transmitted	U32	-
CanRxNum	Number of CAN messages received	U32	-
CanTxOk	CAN transmit percent OK	S32	%
CanRxOk	CAN receive percent OK	S32	%
CanErrorCount	CAN number of errors	U32	-
CanErrorCode	CAN last error code	U16	-
ImuMissedPkts	Number of IMU Packets missed	U32	-
ImuResetCount	Number of IMU Resets detected	U32	-
ImuErrorCount	Number of IMU Errors detected	U32	-
ImuChars	Characters received from the IMU by the navigation computer	U32	-
ImuCharsSkipped	Characters received from the IMU by the navigation computer but not understood by the decoder	U32	-
ImuPkts	Packets received from the IMU by the navigation computer	U32	-
CmdChars	Characters received on this port	U32	-
CmdCharsSkipped	Characters received on this port but not understood by the decoder	U32	-
CmdPkts	Packets received on this port	U32	-
CmdErrors	Errors received on this port	U32	-
TimeMismatch	Time Mismatch Counter	S32	-
SdnSkipCount	Strap down navigator skip count	U32	-
ImuGyroErrorCountX	Number of IMU X gyroscope errors	U32	-
ImuGyroErrorCountY	Number of IMU Y gyroscope errors	U32	-
ImuGyroErrorCountZ	Number of IMU Z gyroscope errors	U32	-
ImuTemp	Average instantaneous IMU temperature	F64	degC

ID: BNS_STATS	Blended navigation statistics source		
ImuAccelSatCount	Number of IMU saturation count	U32	-
ImuGyroSatCount	Number of IMU saturation count	U32	-

Table 87:
Blended navigation statistics source

ID: BNS_ACCDER	Blended navigation system accuracies for derived signals source		
Description	The accuracies for the derived signals of the blended navigation system		
Natural Output Type	OnChange		
Signal ID	Signal Description	Native Type	Native Unit
Speed2dAcc	Combined speed accuracy for horizontal directions	F64	m/s
Speed3dAcc	Combined speed accuracy for all directions	F64	m/s
TrackAcc	Track angle accuracy	F64	deg
SlipAcc	Slip angle accuracy	F64	deg

Table 88:
Blended navigation system accuracies for derived signals source

ID: BNS_BIAS	The biases for different sensors in the blended navigation system source		
Description	The biases of sensors in the blended navigation system		
Natural Output Type	OnChange		
Signal ID	Signal Description	Native Type	Native Unit
WxBias	Gyro Bias X	F64	deg/s
WyBias	Gyro Bias Y	F64	deg/s
WzBias	Gyro Bias Z	F64	deg/s
WxSf	Gyro Scale Factor X	F64	-
WySf	Gyro Scale Factor Y	F64	-
WzSf	Gyro Scale Factor Z	F64	-
AxBias	Accelerometer Bias X	F64	m/s^2
AyBias	Accelerometer Bias Y	F64	m/s^2
AzBias	Accelerometer Bias Z	F64	m/s^2
AxSf	Accelerometer scale factor Xi	F64	-
AySf	Accelerometer scale factor Yi	F64	-
AzSf	Accelerometer scale factor Zi	F64	-

Table 89:
The biases for different sensors in the blended navigation system source

ID: BNS_ACCBIAS	The accuracies of biases of sensors in the blended navigation system		
Description	The accuracies for the biases for different sensors in the blended navigation system source		
Natural Output Type	OnChange		
Signal ID	Signal Description	Native Type	Native Unit
WxBiasAcc	Accuracy of Gyro Bias X	F64	deg/s
WyBiasAcc	Accuracy of Gyro Bias Y	F64	deg/s
WzBiasAcc	Accuracy of Gyro Bias Z	F64	deg/s
WxSfAcc	Accuracy of Gyro Scale Factor X	F64	-
WySfAcc	Accuracy of Gyro Scale Factor Y	F64	-
WzSfAcc	Accuracy of Gyro Scale Factor Z	F64	-
AxBiasAcc	Accuracy of Accelerometer Bias X	F64	m/s^2
AyBiasAcc	Accuracy of Accelerometer Bias Y	F64	m/s^2
AzBiasAcc	Accuracy of Accelerometer Bias Z	F64	m/s^2
AxSfAcc	Accelerometer scale factor accuracy Xi	F64	-
AySfAcc	Accelerometer scale factor accuracy Yi	F64	-
AzSfAcc	Accelerometer scale factor accuracy Zi	F64	-

Table 90:
The accuracies of biases of sensors in the blended navigation system

ID: GNSS_STATS	Statistics for the GNSS engine source		
Description	The statistics from the GNSS engine source		
Natural Output Type	FrequencyBased		
Natural Output Frequency	GNSSRaw		
Signal ID	Signal Description	Native Type	Native Unit
GnssDiffPkts	Differential GNSS packets received	U32	-
GnssVelReject	Number of consecutive GNSS velocity updates rejected	S32	-
GnssAttReject	Number of consecutive GNSS attitude updates rejected	S32	-
GnssDiffChars	Differential GNSS characters received	U32	-
GnssDiffCharsSkipped	Differential GNSS characters skipped	U32	-
GnssPosReject	Number of consecutive GNSS position updates rejected	S32	-

Table 91:
Statistics for the GNSS engine source

ID: BASE_STS	Base station status source		
Description	Signals describing the status of the base station		
Natural Output Type	OnChange		
Signal ID	Signal Description	Native Type	Native Unit
BaseStationID	Differential Station ID	STR	-
BaseStationSlot0	Details of the base station file used for differential corrections	F64	-

Table 92:
Base station status source

ID: BNS_GAD	Generic aiding		
Description	Generic aiding information		
Natural Output Type	OnChange		
Signal ID	Signal Description	Native Type	Native Unit
GADLatestStreamID	Stream ID of the most recently received GAD packet	U8	-
GADNumEarly	Number of too-early packets received on the latest GAD stream	U32	-
GADNumLate	Number of too-late packets received on the latest GAD stream	U32	-
GADNumScheduled	Number of in-time packets received on the latest GAD stream	U32	-
GADLatestStatus	Status of the latest GAD packet received	U8	-
GADChars	The number of characters received on generic aiding streams	U32	-
GADPkts	The number of packets successfully decoded on generic aiding streams	U32	-
GADPosStatus	The GAD position aiding status	U8	-
GADVelStatus	The GAD velocity aiding status	U8	-
GADAttStatus	The GAD attitude aiding status	U8	-
GADAngRateStatus	The GAD angular rate aiding status	U8	-

Table 93:
Generic aiding

ID: BNS_WSP	Wheel speed source		
Description	Signals relating to the wheel speed system		
Natural Output Type	OnChange		
Signal ID	Signal Description	Native Type	Native Unit
WSpeedNano	Wheel speed time of last change	S64	ns
WSpeedFreq	Wheel speed frequency	F64	Hz
WSpeedSpeed	Wheel speed speed	F64	m/s
WSpeedVel	Wheel speed velocity	F64	m/s
WSpeedLeverArmX	Distance to the Wheel Speed Measurement Point in X direction	F64	m
WSpeedLeverArmY	Distance to the Wheel Speed Measurement Point in Y direction	F64	m
WSpeedLeverArmZ	Distance to the Wheel Speed Measurement Point in Z direction	F64	m
WSpeedLeverArmXAcc	Accuracy of Distance to the Wheel Speed Measurement Point in X direction	F64	m
WSpeedLeverArmYAcc	Accuracy off Distance to the Wheel Speed Measurement Point in Y direction	F64	m
WSpeedLeverArmZAcc	Accuracy of Distance to the Wheel Speed Measurement Point in Z direction	F64	m
OptionWSpeedDelay	Wheel speed delay before update	F64	s
OptionWSpeedZVDelay	Wheel speed delay before zero velocity update	F64	s
OptionWSpeedNoiseStd	Wheel speed weight	F64	-
WSpeedScale	Wheelspeed scaling	F64	1/m
WSpeedScaleStd	Wheelspeed scaling accuracy	F64	%
WSpeedTimeUnchanged	Time since the wheelspeed count changed	F64	s
WSpeedCount	Cyclic wheelspeed input counts	F64	-

Table 94:
Wheel speed source

ID: BNS_CAM	Camera source		
Description	The camera source		
Natural Output Type	OnTrigger		
Signal ID	Signal Description	Native Type	Native Unit
DigitalOut2UpdateCount	Camera 2 output update count	S32	-
DigitalOutUpdateCount	Camera output/distance output update count	S32	-

Table 95:
Camera source

ID: GNSS_ACC	GNSS engine accuracies		
Description	The accuracies from the GNSS engine		
Natural Output Type	OnChange		
Signal ID	Signal Description	Native Type	Native Unit
PDOP	PDOP of the GNSS receiver being used for position	F64	-
HDOP	HDOP of the GNSS receiver being used for position	F64	-
VDOP	VDOP of the GNSS receiver being used for position	F64	-

Table 96:
GNSS engine accuracies

ID: GNSS_EGC	External GNSS card source		
Description	The source containing the external GNSS card signals		
Natural Output Type	FrequencyBased		
Natural Output Frequency	GNSSRaw		
Signal ID	Signal Description	Native Type	Native Unit
GnssExt1_AntStatus	GNSS Antenna Status	U8	-
GnssExt1_AntPower	GNSS Antenna Power Status	S8	-
GnssExt1_NumSats	Number of Satellites Tracked	U8	-
GnssExt1_Chars	Characters received from the External GNSS by the navigation computer	U32	-
GnssExt1_CharsSkipped	Characters received from the External GNSS by the navigation computer but not understood by the decoder	U32	-
GnssExt1_Pkts	Packets received from External GNSS by the navigation computer	U32	-
GnssExt1_OldPkts	Packets received from External GNSS by the navigation computer that could not be used to update the Kalman filter (e.g too old)	U32	-
GnssExt1_NumGpsL1	External1 receiver GPS L1 number of raw measurements received	U8	-
GnssExt1_NumGpsL2	External1 receiver GPS L2 number of raw measurements received	U8	-
GnssExt1_NumGlonassL1	External1 receiver GLONASS L1 number of raw measurements received	U8	-
GnssExt1_NumGlonassL2	External1 receiver GLONASS L2 number of raw measurements received	U8	-
GnssExt1_NumGalileoE1	External1 receiver Galileo E1 number of raw measurements received	U8	-
GnssExt1_NumGalileoE5b	External1 receiver Galileo E5b number of raw measurements received	U8	-
GnssExt1_NumBeiDou2B1	External1 receiver BeiDou Gen 2 B1 number of raw measurements received	U8	-
GnssExt1_NumBeiDou2B2	External1 receiver BeiDou Gen 2 B2 number of raw measurements received	U8	-

ID: GNSS_EGC	External GNSS card source		
GnssExt1_NumBeiDou3B1	External1 receiver BeiDou Gen 3 B1 number of raw measurements received	U8	-

Table 97:
External GNSS card source

Renamed Signals

Old	New
ASP1	MeasPt1_Slip
ASP2	MeasPt2_Slip
ASP3	MeasPt3_Slip
ASP4	MeasPt4_Slip
ASP5	MeasPt5_Slip
ASP6	MeasPt6_Slip
ASP7	MeasPt7_Slip
ASP8	MeasPt8_Slip
ATP1	MeasPt1_Track
ATP2	MeasPt2_Track
ATP3	MeasPt3_Track
ATP4	MeasPt4_Track
ATP5	MeasPt5_Track
ATP6	MeasPt6_Track
ATP7	MeasPt7_Track
ATP8	MeasPt8_Track
AhX	IsoAhX
AhY	IsoAhY
AhZ	IsoAhZ
Ahd	Ad
Ahf	Af
Ahl	Al
AnX	IsoAnX
AnY	IsoAnY
AnZ	IsoAnZ
AngGrad	AngleGradient

Old	New
AoX	IsoAoX
AoY	IsoAoY
AoZ	IsoAoZ
CP1	MeasPt1_Curvature
CP2	MeasPt2_Curvature
CP3	MeasPt3_Curvature
CP4	MeasPt4_Curvature
CP5	MeasPt5_Curvature
CP6	MeasPt6_Curvature
CP7	MeasPt7_Curvature
CP8	MeasPt8_Curvature
Crv	Curvature
GpsPosMode	GnssPosMode
GpsVelMode	GnssVelMode
GpsAttMode	GnssAttMode
DiffAge	GnssDiffAge
GpsNumObs	GnssPosNumSats
HeaSatUsed	GnssAttNumSats
Dist2D	Dist2d
Dist2DHold	Dist2dHold
Dist3D	Dist3d
Dist3DHold	Dist3dHold
DownAcc	AltAcc
FiltAhX	FiltIsoAhX
FiltAhY	FiltIsoAhY
FiltAhZ	FiltIsoAhZ
FiltAhd	FiltAd
FiltAhf	FiltAf

Old	New
FiltAhl	FiltAI
FiltWhd	FiltWd
FiltWhf	FiltWf
FiltWhl	FiltWI
FiltAnX	FiltIsoAnX
FiltAnY	FiltIsoAnY
FiltAnZ	FiltIsoAnZ
FiltAoX	FiltIsoAoX
FiltAoY	FiltIsoAoY
FiltAoZ	FiltIsoAoZ
FiltJhX	FiltIsoJhX
FiltJhY	FiltIsoJhY
FiltJhZ	FiltIsoJhZ
FiltJhd	FiltJd
FiltJhf	FiltJf
FiltJhl	FiltJI
FiltJnX	FiltIsoJnX
FiltJnY	FiltIsoJnY
FiltJnZ	FiltIsoJnZ
FiltJoX	FiltIsoJoX
FiltJoY	FiltIsoJoY
FiltJoZ	FiltIsoJoZ
FiltYhX	FiltIsoYhX
FiltYhY	FiltIsoYhY
FiltYhZ	FiltIsoYhZ
FiltYhd	FiltYd
FiltYhf	FiltYf
FiltYhl	FiltYI

Old	New
FiltYnX	FiltIsoYnX
FiltYnY	FiltIsoYnY
FiltYnZ	FiltIsoYnZ
FiltYoX	FiltIsoYoX
FiltYoY	FiltIsoYoY
FiltYoZ	FiltIsoYoZ
GNSST	Nano
IPitch	IsoPitch
IRoll	IsoRoll
IYaw	IsoYaw
JhX	IsoJhX
JhY	IsoJhY
JhZ	IsoJhZ
Jhd	Jd
Jhf	Jf
Jhl	Jl
JnX	IsoJnX
JnY	IsoJnY
JnZ	IsoJnZ
JoX	IsoJoX
JoY	IsoJoY
JoZ	IsoJoZ
LocalTrAg	RefFrameTrack
LocalVx	RefFrameVelX
LocalVy	RefFrameVelY
LocalX	RefFrameX
LocalY	RefFrameY
LocalYaw	RefFrameYaw

Old	New
NavStatus	InsNavMode
PosMode	GpsPosMode
SatUsed	GpsNumObs
SIAg	Slip
Speed2D	Speed2d
Speed3D	Speed3d
SrNo	SerialNumber
TrAg	Track
VelMode	GpsVelMode
VhX	IsoVhX
VhY	IsoVhY
VhZ	IsoVhZ
Vhf	Vf
Vhl	VI
Vhd	Vd
VnX	IsoVnX
VnY	IsoVnY
VnZ	IsoVnZ
VoX	IsoVoX
VoY	IsoVoY
VoZ	IsoVoZ
WhX	IsoWhX
WhY	IsoWhY
WhZ	IsoWhZ
Whd	Wd
Whf	Wf
Whl	WI
WnX	IsoWnX

Old	New
WnY	IsoWnY
WnZ	IsoWnZ
WoX	IsoWoX
WoY	IsoWoY
WoZ	IsoWoZ
YhX	IsoYhX
YhY	IsoYhY
YhZ	IsoYhZ
Yhd	Yd
Yhf	Yf
Yhl	Yl
YnX	IsoYnX
YnY	IsoYnY
YnZ	IsoYnZ
YoX	IsoYoX
YoY	IsoYoY
YoZ	IsoYoZ
IYawAcc	IsoYawAcc
IPitchAcc	IsoPitchAcc
IRollAcc	IsoRollAcc
VnXAcc	IsoVnXAcc
VnYAcc	IsoVnYAcc
VnZAcc	IsoVnZAcc
Ahf1	MeasPt1_Af
Ahl1	MeasPt1_Al
Ahd1	MeasPt1_Ad
Vhf1	MeasPt1_Vf
Vhl1	MeasPt1_Vl

Old	New
Vhd1	MeasPt1_Vd
Ahf2	MeasPt2_Af
Ahl2	MeasPt2_AI
Ahd2	MeasPt2_Ad
Vhf2	MeasPt2_Vf
Vhl2	MeasPt2_VI
Vhd2	MeasPt2_Vd
Ahf3	MeasPt3_Af
Ahl3	MeasPt3_AI
Ahd3	MeasPt3_Ad
Vhf3	MeasPt3_Vf
Vhl3	MeasPt3_VI
Vhd3	MeasPt3_Vd
Ahf4	MeasPt4_Af
Ahl4	MeasPt4_AI
Ahd4	MeasPt4_Ad
Vhf4	MeasPt4_Vf
Vhl4	MeasPt4_VI
Vhd4	MeasPt4_Vd
Ahf5	MeasPt5_Af
Ahl5	MeasPt5_AI
Ahd5	MeasPt5_Ad
Vhf5	MeasPt5_Vf
Vhl5	MeasPt5_VI
Vhd5	MeasPt5_Vd
Ahf6	MeasPt6_Af
Ahl6	MeasPt6_AI
Ahd6	MeasPt6_Ad

Old	New
Vhf6	MeasPt6_Vf
Vhl6	MeasPt6_Vl
Vhd6	MeasPt6_Vd
Ahf7	MeasPt7_Af
Ahl7	MeasPt7_Al
Ahd7	MeasPt7_Ad
Vhf7	MeasPt7_Vf
Vhl7	MeasPt7_Vl
Vhd7	MeasPt7_Vd
Ahf8	MeasPt8_Af
Ahl8	MeasPt8_Al
Ahd8	MeasPt8_Ad
Vhf8	MeasPt8_Vf
Vhl8	MeasPt8_Vl
Vhd8	MeasPt8_Vd
An1	MeasPt1_An
An2	MeasPt2_An
An3	MeasPt3_An
An4	MeasPt4_An
An5	MeasPt5_An
An6	MeasPt6_An
An7	MeasPt7_An
Ae1	MeasPt1_Ae
Ae2	MeasPt2_Ae
Ae3	MeasPt3_Ae
Ae4	MeasPt4_Ae
Ae5	MeasPt5_Ae
Ae6	MeasPt6_Ae

Old	New
Ae7	MeasPt7_Ae
Ae8	MeasPt8_Ae
An8	MeasPt8_An
Ad1	MeasPt1_Ad
Ad2	MeasPt2_Ad
Ad3	MeasPt3_Ad
Ad4	MeasPt4_Ad
Ad5	MeasPt5_Ad
Ad6	MeasPt6_Ad
Ad7	MeasPt7_Ad
Ad8	MeasPt8_Ad
S2D1	MeasPt1_Speed2d
S2D2	MeasPt2_Speed2d
S2D3	MeasPt3_Speed2d
S2D4	MeasPt4_Speed2d
S2D5	MeasPt5_Speed2d
S2D6	MeasPt6_Speed2d
S2D7	MeasPt7_Speed2d
S2D8	MeasPt8_Speed2d

Table 98:
Old to new signal ID renames

Revision history

Revision	Comment
250716	Initial draft.
260707	Updates for UCOM Version 1 including new messages, signals, and output conditions.

Table 99:
Revision history

Appendix A

CRC-32 Look-up table values:

0x00000000, 0x77073096, 0xEE0E612C, 0x990951BA, 0x076DC419, 0x706AF48F, 0xE963A535,
0x9E6495A3, 0x0EDB8832, 0x79DCB8A4, 0xE0D5E91E, 0x97D2D988, 0x09B64C2B, 0x7EB17CBD,
0xE7B82D07, 0x90BF1D91, 0x1DB71064, 0x6AB020F2, 0xF3B97148, 0x84BE41DE, 0x1ADAD47D,
0x6DDDE4EB, 0xF4D4B551, 0x83D385C7, 0x136C9856, 0x646BA8C0, 0xFD62F97A, 0x8A65C9EC,
0x14015C4F, 0x63066CD9, 0xFA0F3D63, 0x8D080DF5, 0x3B6E20C8, 0x4C69105E, 0xD56041E4,
0xA2677172, 0x3C03E4D1, 0x4B04D447, 0xD20D85FD, 0xA50AB56B, 0x35B5A8FA, 0x42B2986C,
0xDBBBC9D6, 0xACBCF940, 0x32D86CE3, 0x45DF5C75, 0xDCD60DCF, 0xABD13D59, 0x26D930AC,
0x51DE003A, 0xC8D75180, 0xBF06116, 0x21B4F4B5, 0x56B3C423, 0xCFBA9599, 0xB8BDA50F,
0x2802B89E, 0x5F058808, 0xC60CD9B2, 0xB10BE924, 0x2F6F7C87, 0x58684C11, 0xC1611DAB,
0xB6662D3D, 0x76DC4190, 0x01DB7106, 0x98D220BC, 0xEFD5102A, 0x71B18589, 0x06B6B51F,
0x9FBFE4A5, 0xE8B8D433, 0x7807C9A2, 0xF00F934, 0x9609A88E, 0xE10E9818, 0x7F6A0DBB,
0x086D3D2D, 0x91646C97, 0xE6635C01, 0x6B6B51F4, 0x1C6C6162, 0x856530D8, 0xF262004E,
0x6C0695ED, 0x1B01A57B, 0x8208F4C1, 0xF50FC457, 0x65B0D9C6, 0x12B7E950, 0x8BBEB8EA,
0xFCB9887C, 0x62DD1DDF, 0x15DA2D49, 0x8CD37CF3, 0xFBD44C65, 0x4DB26158, 0x3AB551CE,
0xA3BC0074, 0xD4BB30E2, 0x4ADFA541, 0x3DD895D7, 0xA4D1C46D, 0xD3D6F4FB, 0x4369E96A,
0x346ED9FC, 0xAD678846, 0xDA60B8D0, 0x44042D73, 0x33031DE5, 0xAA0A4C5F, 0xDD0D7CC9,
0x5005713C, 0x270241AA, 0xBE0B1010, 0xC90C2086, 0x5768B525, 0x206F85B3, 0xB966D409,
0xCE61E49F, 0x5EDEF90E, 0x29D9C998, 0xB0D09822, 0xC7D7A8B4, 0x59B33D17, 0x2EB40D81,
0xB7BD5C3B, 0xC0BA6CAD, 0xEDB88320, 0x9ABFB3B6, 0x03B6E20C, 0x74B1D29A, 0xEAD54739,
0x9DD277AF, 0x04DB2615, 0x73DC1683, 0xE3630B12, 0x94643B84, 0x0D6D6A3E, 0x7A6A5AA8,
0xE40ECF0B, 0x9309FF9D, 0x0A00AE27, 0x7D079EB1, 0xF00F9344, 0x8708A3D2, 0x1E01F268,
0x6906C2FE, 0xF762575D, 0x806567CB, 0x196C3671, 0x6E6B06E7, 0xFED41B76, 0x89D32BE0,
0x10DA7A5A, 0x67DD4ACC, 0xF9B9DF6F, 0x8EBEFFF9, 0x17B7BE43, 0x60B08ED5, 0xD6D6A3E8,
0xA1D1937E, 0x38D8C2C4, 0x4FDDF252, 0xD1BB67F1, 0xA6BC5767, 0x3FB506DD, 0x48B2364B,
0xD80D2BDA, 0xAF0A1B4C, 0x36034AF6, 0x41047A60, 0xDF60EFC3, 0xA867DF55, 0x316E8EEF,
0x4669BE79, 0xCB61B38C, 0xBC66831A, 0x256FD2A0, 0x5268E236, 0xCC0C7795, 0xBB0B4703,
0x220216B9, 0x5505262F, 0xC5BA3BBE, 0xB2BD0B28, 0x2BB45A92, 0x5CB36A04, 0xC2D7FFA7,
0xB5D0CF31, 0x2CD99E8B, 0x5BDEAE1D, 0x9B64C2B0, 0xEC63F226, 0x756AA39C, 0x026D930A,
0x9C0906A9, 0xEB0E363F, 0x72076785, 0x05005713, 0x95BF4A82, 0xE2B87A14, 0x7BB12BAE,
0x0CB61B38, 0x92D28E9B, 0xE5D5BE0D, 0x7CDEFB7, 0xBDBDF21, 0x86D3D2D4, 0xF1D4E242,
0x68DD3F8, 0x1FDA836E, 0x81BE16CD, 0xF6B9265B, 0x6FB077E1, 0x18B74777, 0x88085AE6,
0xFF0F6A70, 0x66063BCA, 0x11010B5C, 0x8F659EFF, 0xF862AE69, 0x616BFFD3, 0x166CCF45,
0xA00AE278, 0xD70DD2EE, 0x4E048354, 0x3903B3C2, 0xA7672661, 0xD06016F7, 0x4969474D,
0x3E6E77DB, 0xAED16A4A, 0xD9D65ADC, 0x40DF0B66, 0x37D83BF0, 0xA9BCAE53, 0xDEBB9EC5,
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0xB40BBE37, 0xC30C8EA1, 0x5A05DF1B, 0x2D02EF8D

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